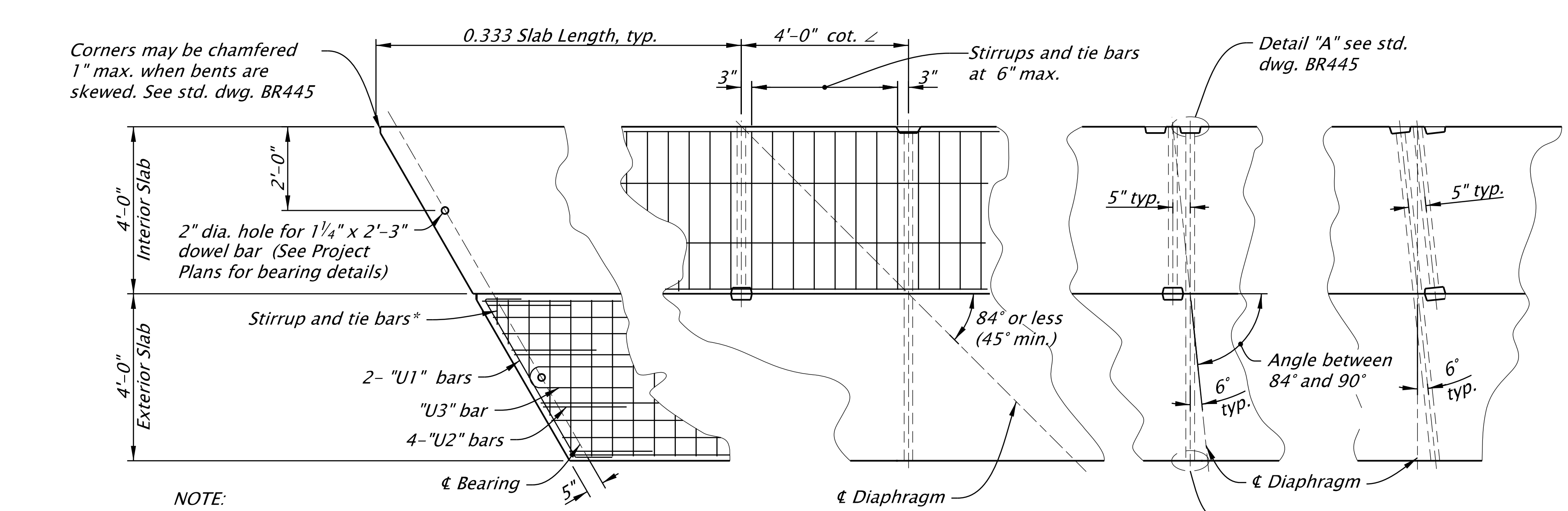
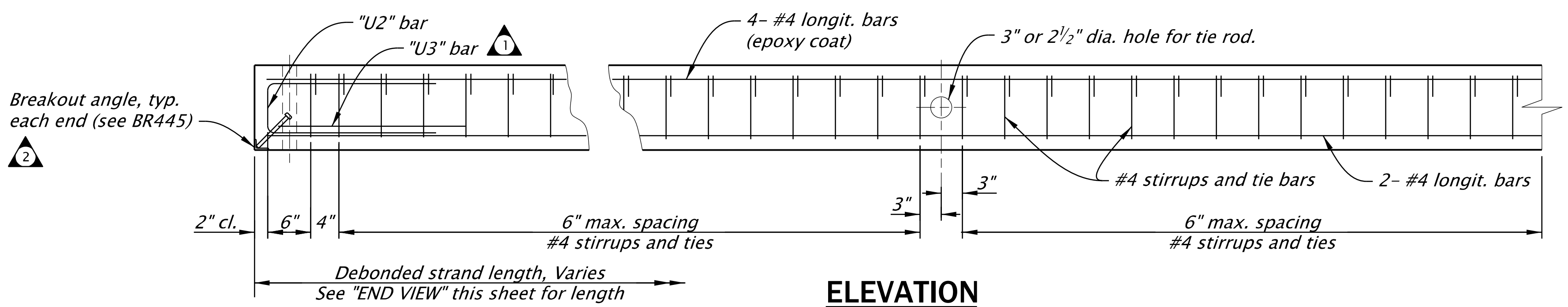


PLAN
Scale: ¼" = 1'-0"



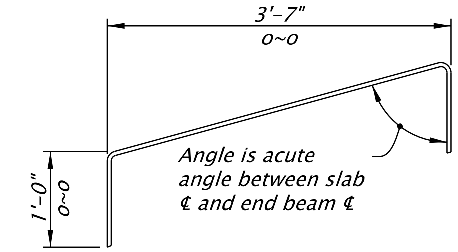
ELEVATION
Scale: ½" =1'-0"

NOTE "A" (Slab End Bars)
2- #4 x 7'-6", slabs 6, 7 and 8.
Place bars each end of each slab (Epoxy coat).

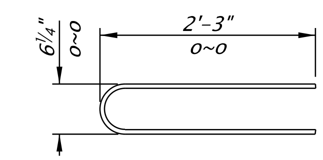
NOTE:
Grout keyway as specified in General Notes.
Omit keyway on exterior side of exterior slabs.
Keyway is continuous.

SECTION PROPERTIES

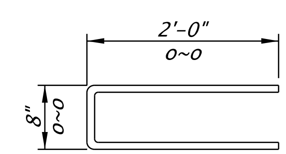
Area =	568 in²
c.g. =	5.97 in
I =	6,851 in⁴
St =	1137 in³
Sb =	1147 in³
Weight =	612 lbs/ft
J =	23,400 in⁴
K =	0.59
V/S =	4.74



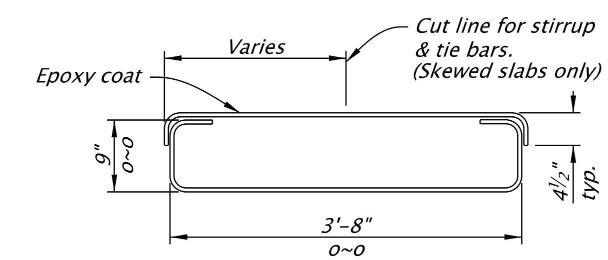
#4 "U1" BAR
(4- required per slab)



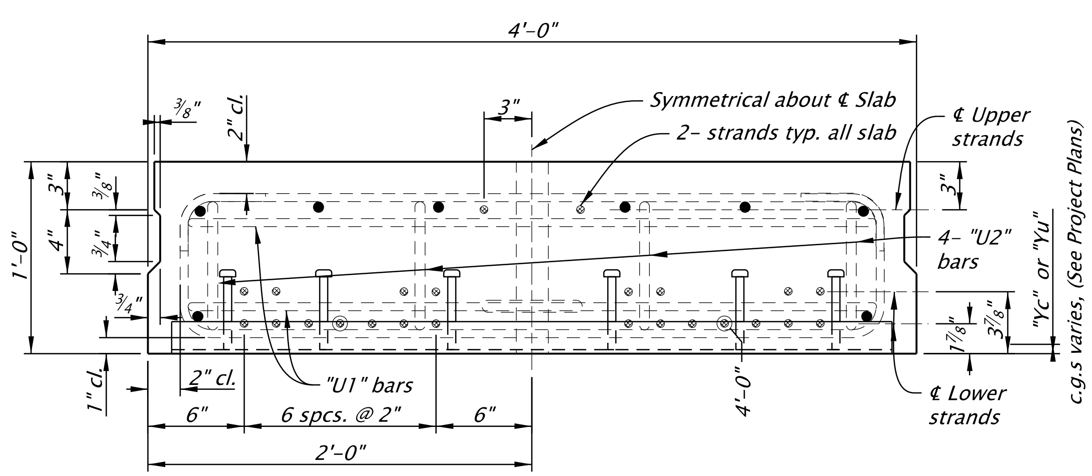
#6 "U3" BAR (A706)
(2- required per slab)



#5 "U2" BAR
(8- required per slab)

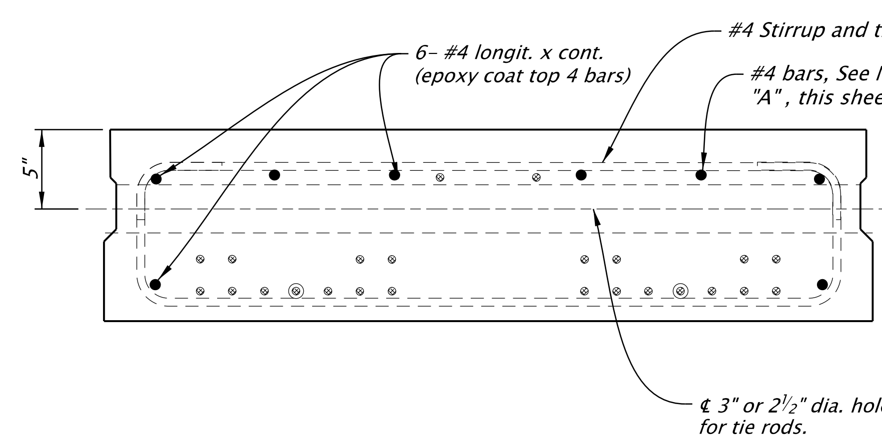


#4 STIRRUP & TIE



END VIEW
Scale: 1" = 1'-0"

DEBONDED STRANDS:
Dimension shown indicates the length of tube required, at both ends, to debond the indicated strand. See the "PRECAST PRESTRESSED SLAB SCHEDULE" in the project plans for the strand pattern detail.
Detension strands in order of increasing tube length shown.



MIDSPAN VIEW
Scale: 1" = 1'-0"

Accompanied by dwg. BR445

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without first consulting a Registered Professional Engineer.

All materials shall be in accordance with the current Oregon Standard Specifications.

OREGON STANDARD DRAWINGS

12" PRECAST PRESTRESSED SLAB

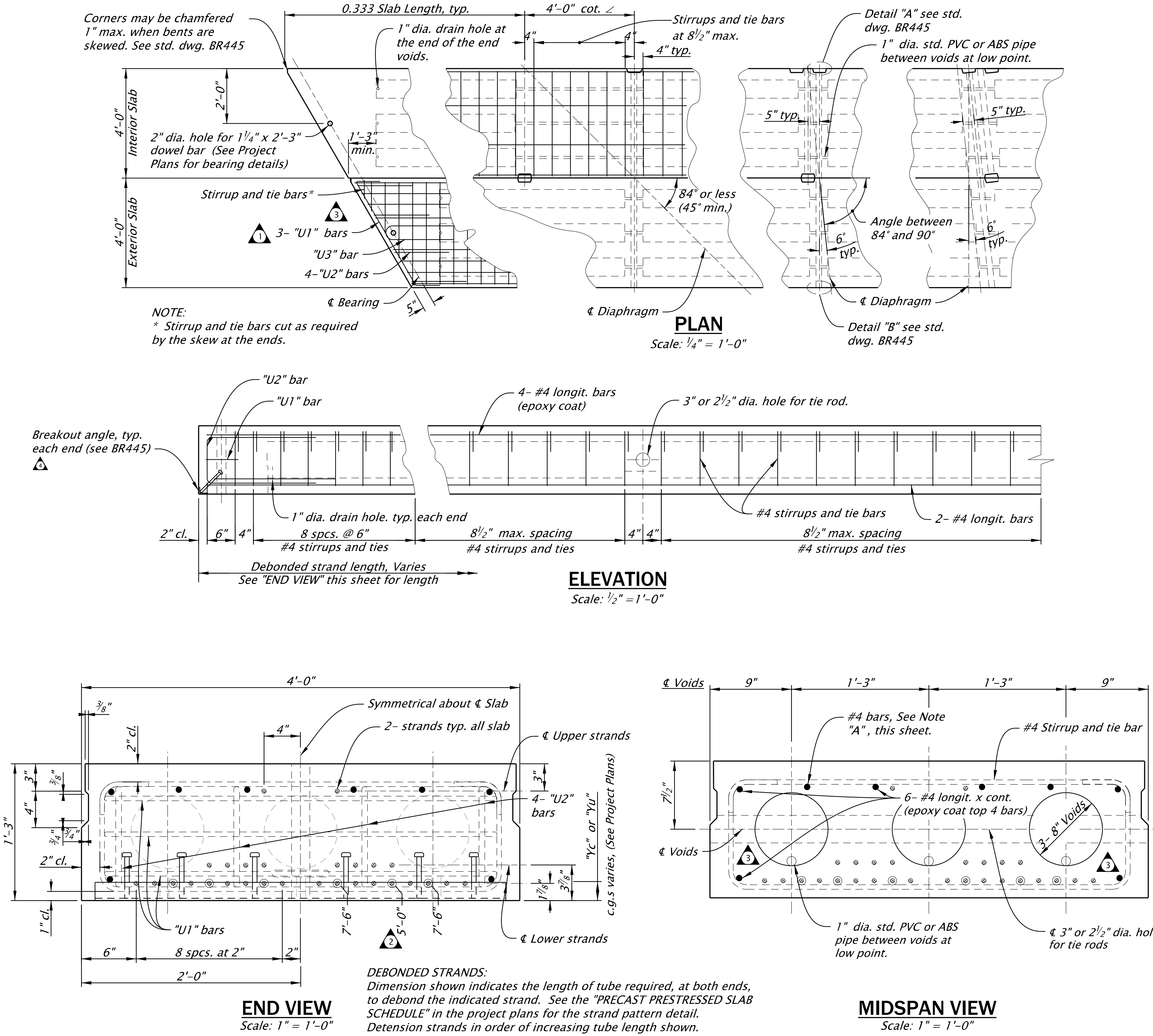
2024

	DATE	REVISION DESCRIPTION
1	07-2020	Revised stirrup and tie bar spacing
	07-2020	Updated drawing to current stds.
2	07-2022	Added breakout angle

CALC. BOOK NO. N/A SDR DATE 08-JULY-2022 BR400

JUL-2022

BR405.dgn



DEBONDED STRANDS:
Dimension shown indicates the length of tube required, at both ends, to debond the indicated strand. See the "PRECAST PRESTRESSED SLAB SCHEDULE" in the project plans for the strand pattern detail. Detension strands in order of increasing tube length shown.

NOTE "A" (Slab End Bars)
2- #4 x 7'-6", slab 5
2- #4 x 10'-0", slabs 6, 7, and 8
2- #4 x 12'-6", slab 9
Place bars each end of each slab (Epoxy coat).

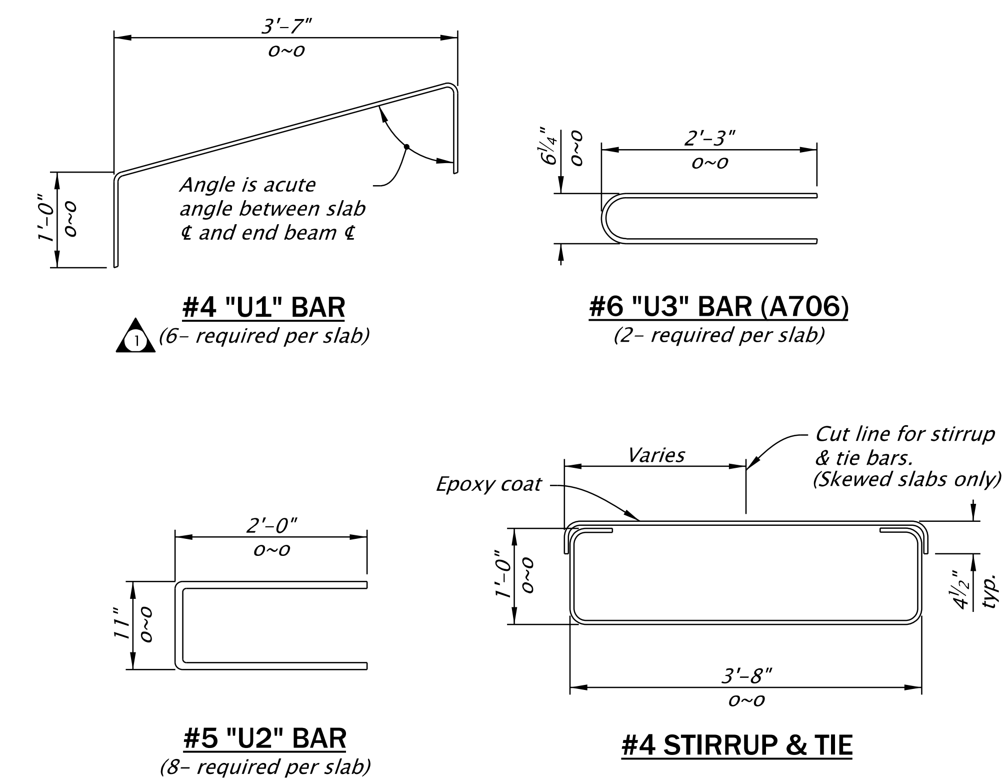
NOTE:
Grout keyway as specified in General Notes.
Omit keyway on exterior side of exterior slabs.
Keyway is continuous.

SECTION PROPERTIES

Area =	562in²
c.g. =	7.45 in
I =	12,769 in⁴
St =	1691in³
Sb =	1714 in³
Weight =	605 lbs/ft
J =	31,900 in⁴
K =	0.69
V/S =	4.46
Form wt = (tubes)	10 lbs/ft
Total wt w/forms=	615 lbs/ft

Diaphragm Weight

No Skew	170 lb
15° Skew	270 lb
30° Skew	470 lb
45° Skew	750 lb



Accompanied by dwg. BR445

The selection and use of this Standard Drawing, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user and should not be used without first consulting a Registered Professional Engineer.

All materials shall be in accordance with the current Oregon Standard Specifications.

OREGON STANDARD DRAWINGS

15" PRECAST PRESTRESSED SLAB

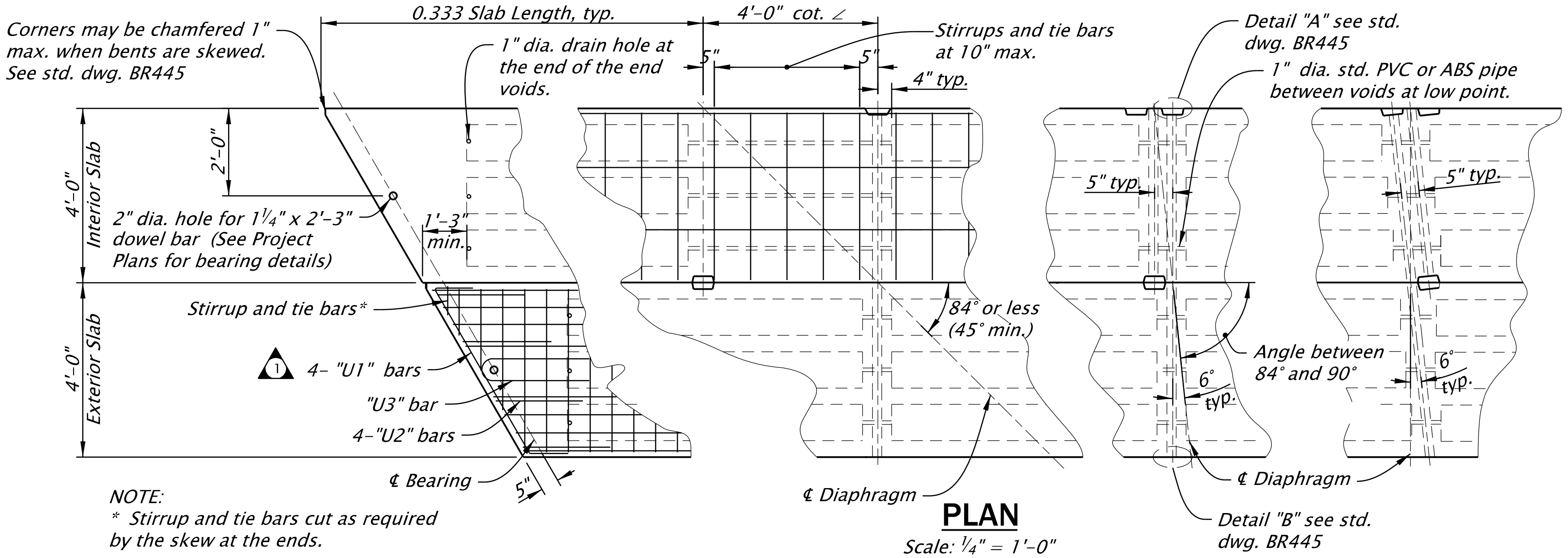
2024

	DATE	REVISION DESCRIPTION
4	07-2020	Added end zone reinf.
2	07-2020	Revised debonded lengths
	07-2020	Updated drawing to current stds.
3	01-2022	Revised #4 bars to be inside of #4 stirrup, Removed "D1" hoop notation.
4	07-2022	Added breakout angle

CALC. BOOK NO. N/A | SDR DATE 08-JUL-2022 | BR405

JUL-2022

BR410.dgn



PLAN
Scale: ¼" = 1'-0"

NOTE "A" (Slab End Bars)
2- #4 x 10'-0", slabs 4, 5, and 6
4- #4 x 15'-0", slabs 7, 8 and 9
Place bars each end of each slab (Epoxy coat).

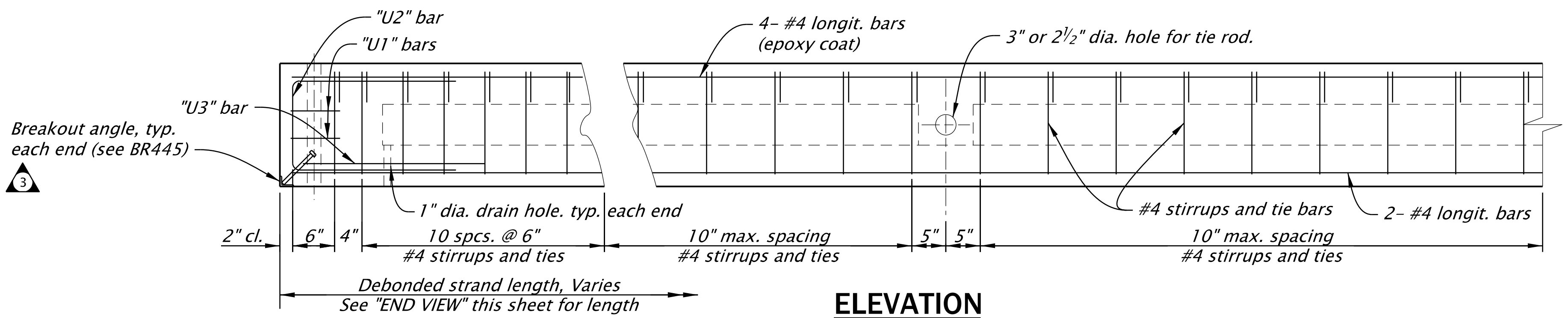
NOTE:
Grout keyway as specified in General Notes.
Omit keyway on exterior side of exterior slabs.
Keyway is continuous.

SECTION PROPERTIES

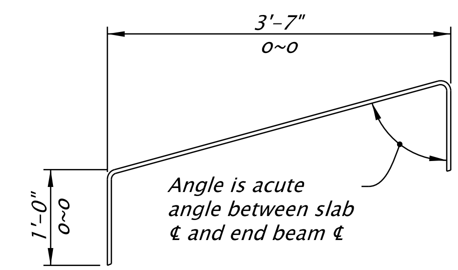
Area =	621in²
c.g. =	8.94 in
I =	21,626 in⁴
St =	2386 in³
Sb =	2420 in³
Weight =	668 lbs/ft
J =	52,200 in⁴
K =	0.71
V/S =	4.70
Form wt = (tubes)	14 lbs/ft
Total wt w/forms=	682 lbs/ft

Diaphragm Weight

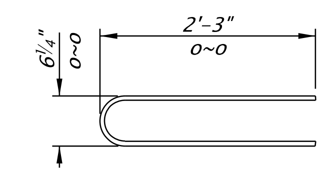
No Skew	260 lb
15° Skew	430 lb
30° Skew	740 lb
45° Skew	1170 lb



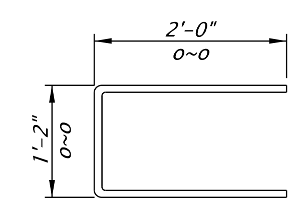
ELEVATION
Scale: ½" =1'-0"



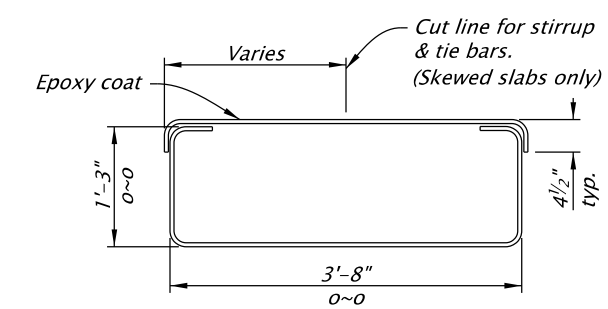
#4 "U1" BAR
(8- required per slab)



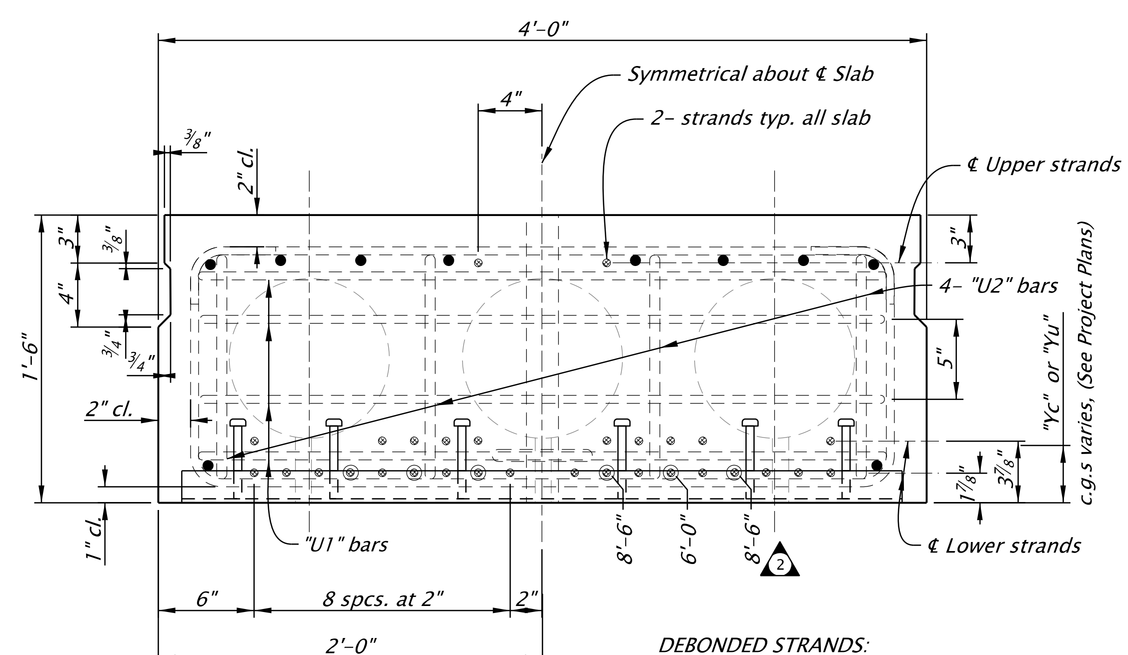
#6 "U3" BAR (A706)
(2- required per slab)



#5 "U2" BAR
(8- required per slab)

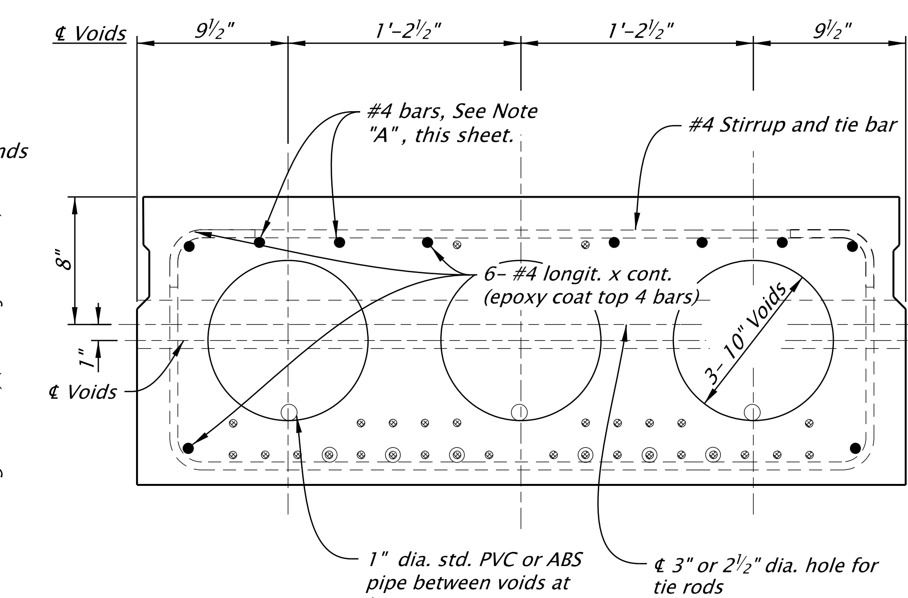


#4 STIRRUP & TIE



END VIEW
Scale: 1" = 1'-0"

DEBONDED STRANDS:
Dimension shown indicates the length of tube required, at both ends, to debond the indicated strand. See the "PRECAST PRESTRESSED SLAB SCHEDULE" in the project plans for the strand pattern detail.
Detension strands in order of increasing tube length shown.



MIDSPAN VIEW
Scale: 1" = 1'-0"

Accompanied by dwg. BR445

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All materials shall be in accordance with the current Oregon Standard Specifications.

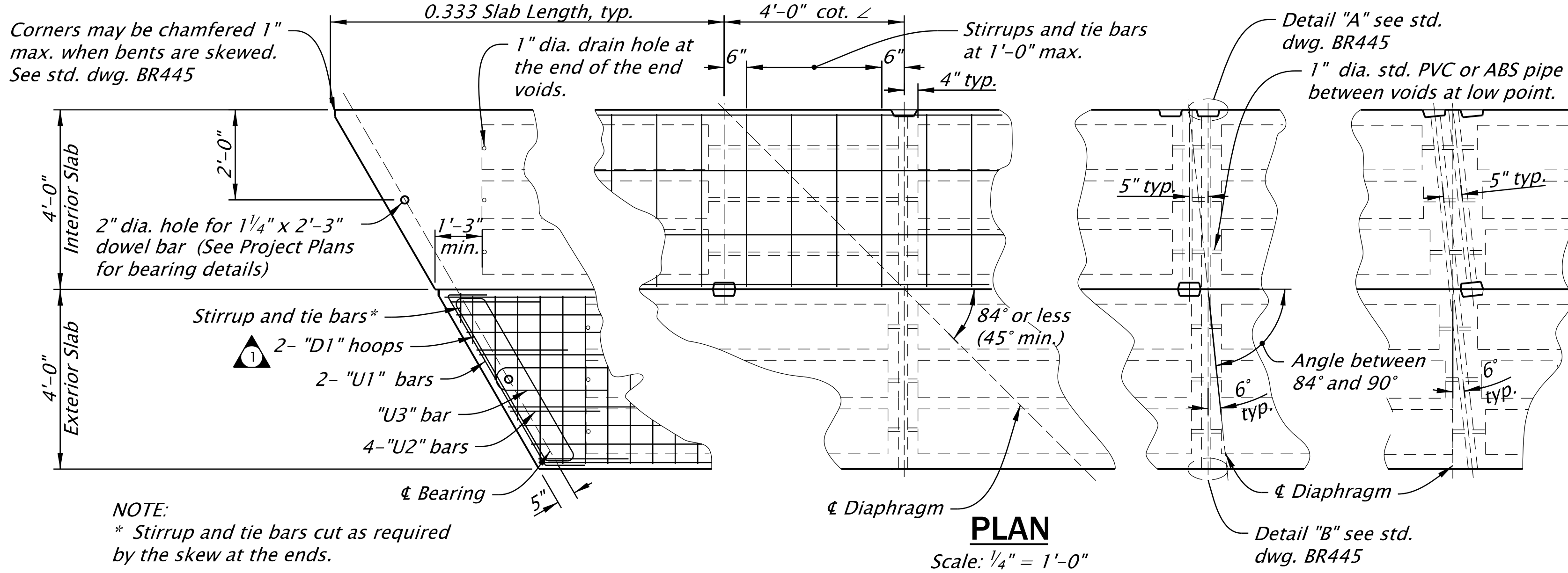
OREGON STANDARD DRAWINGS

18" PRECAST PRESTRESSED SLAB

2024

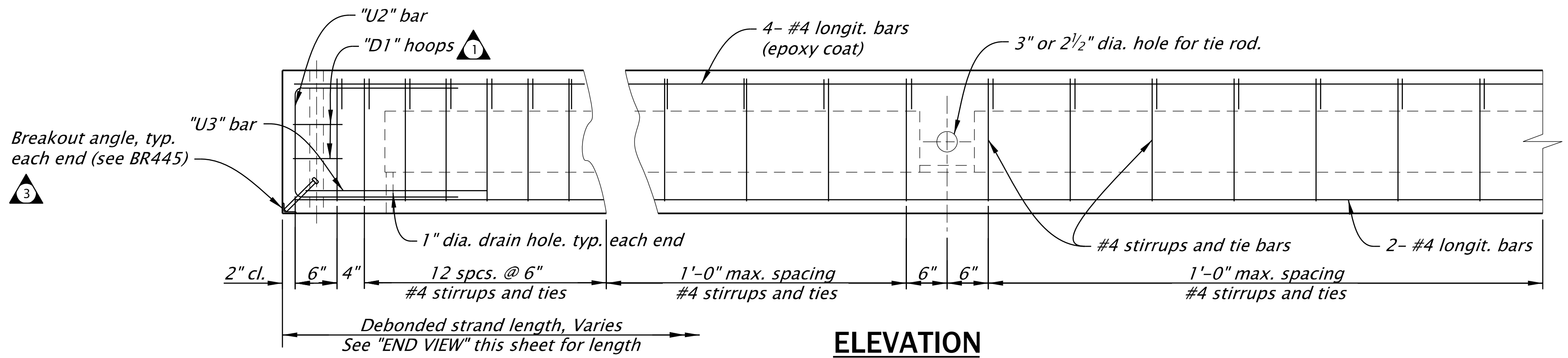
DATE	REVISION DESCRIPTION
1 07-2020	Added end zone reinf.
2 07-2020	Revised debonded lengths
07-2020	Updated drawing to current stds.
3 07-2022	Added breakout angle.

CALC. BOOK NO. N/A | SDR DATE 08-JUL/2022 | BR410



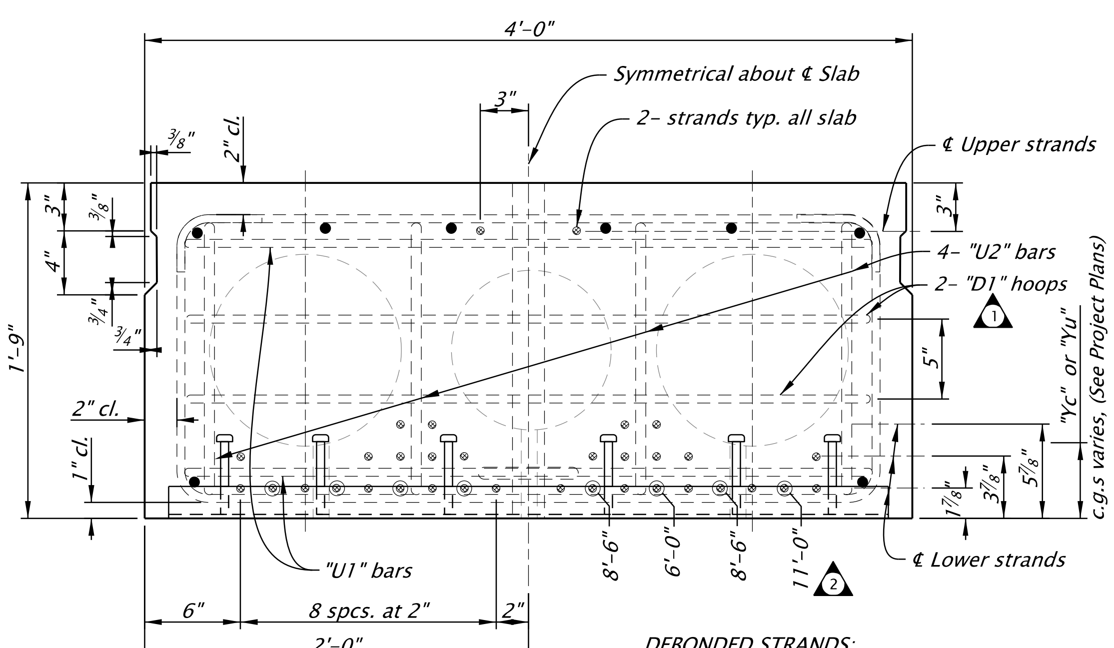
PLAN

Scale: 1/4" = 1'-0"



ELEVATION

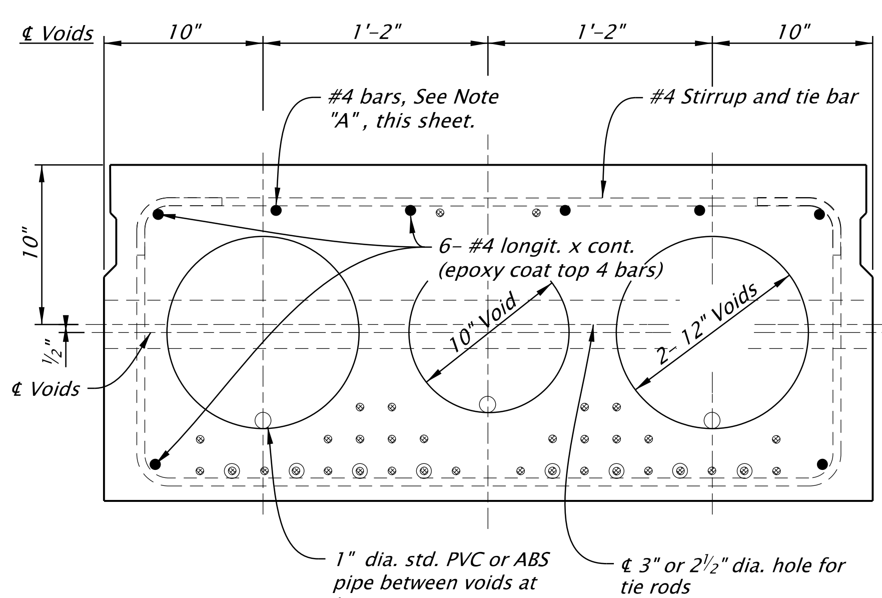
Scale: 1/2" = 1'-0"



END VIEW

Scale: 1" = 1'-0"

DEBONDED STRANDS:
Dimension shown indicates the length of tube required, at both ends, to debond the indicated strand. See the "PRECAST PRESTRESSED SLAB SCHEDULE" in the project plans for the strand pattern detail. Detension strands in order of increasing tube length shown.



MIDSPAN VIEW

Scale: 1" = 1'-0"

NOTE "A" (Slab End Bars)
2- #4 x 10'-0", slabs 6 and 7
2- #4 x12'-6", slabs 8 and 9
2- #4 x 15'-0", slabs10 and 11
Place bars each end of each slab (Epoxy coat).

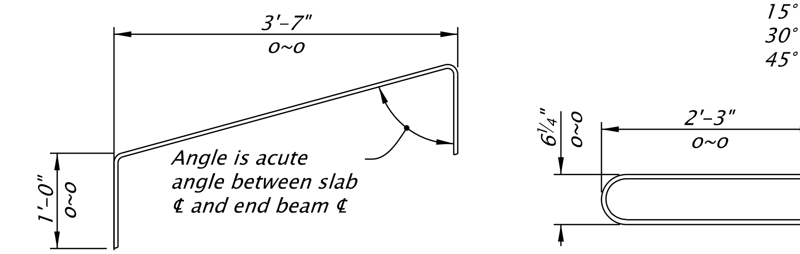
NOTE:
Grout keyway as specified in General Notes.
Omit keyway on exterior side of exterior slabs.
Keyway is continuous.

SECTION PROPERTIES

Area =	696in²
c.g. =	10.43 in
I =	34,152 in⁴
St =	3230in³
Sb =	3275in³
Weight =	749 lbs/ft
J =	77,100 in⁴
K =	0.73
V/S =	5.04
Form wt = (tubes)	16 lbs/ft
Total wt w/forms=	765 lbs/ft

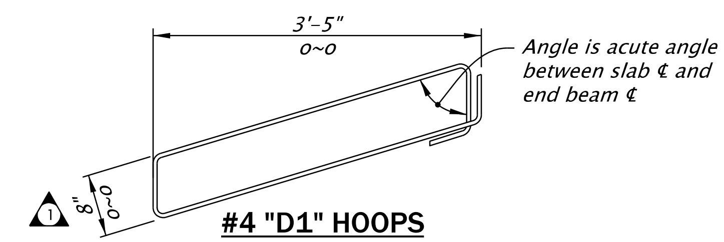
Diaphragm Weight

No Skew	340 lb
15° Skew	560 lb
30° Skew	960 lb
45° Skew	1520 lb

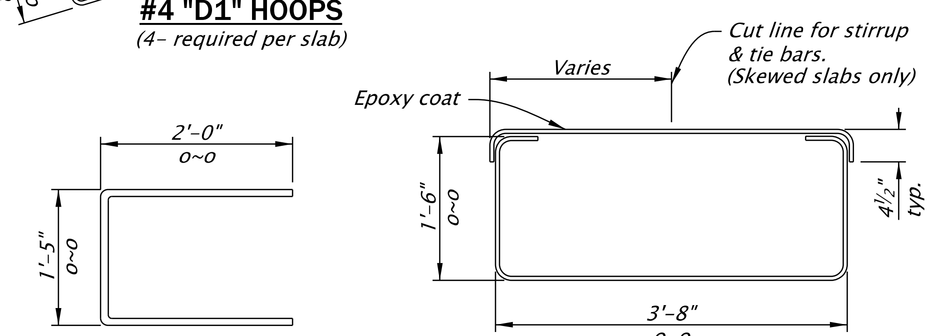


#4 "U1" BAR
(4- required per slab)

#6 "U3" BAR (A706)
(2- required per slab)



#4 "D1" HOOPS
(4- required per slab)



#5 "U2" BAR
(8- required per slab)

#4 STIRRUP & TIE

Accompanied by dwg. BR445

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All materials shall be in accordance with the current Oregon Standard Specifications.

OREGON STANDARD DRAWINGS

21" PRECAST PRESTRESSED SLAB

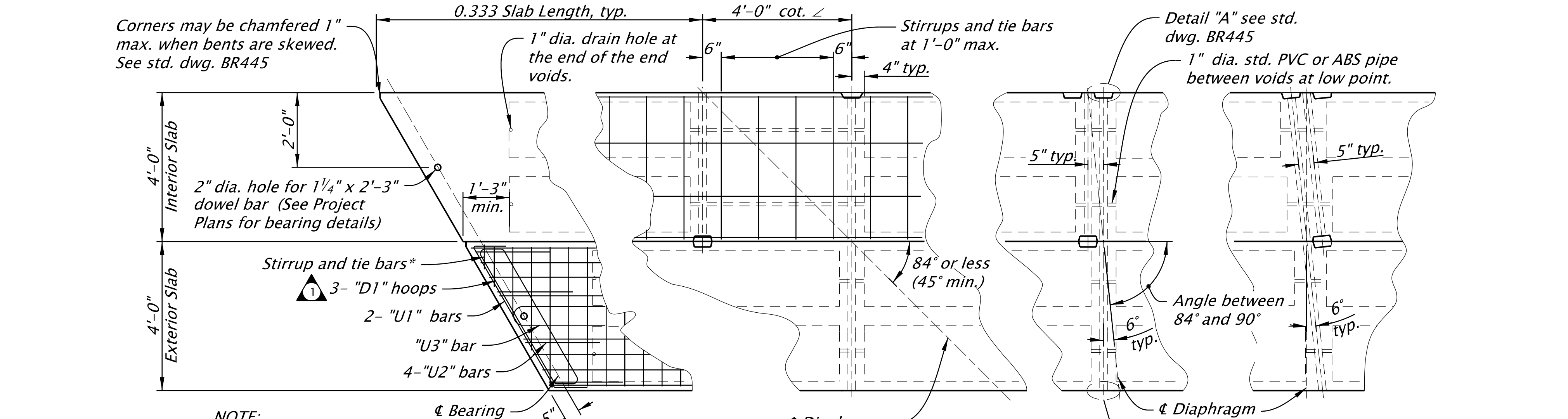
2024

DATE	REVISION DESCRIPTION
07-2020	Added end zone reinf.
07-2020	Revised debonded length.
07-2020	Updated drawing to current stds.
07-2022	Added breakout angle.

CALC. BOOK NO. N/A | SDR DATE 08-JUL-2022 | BR415

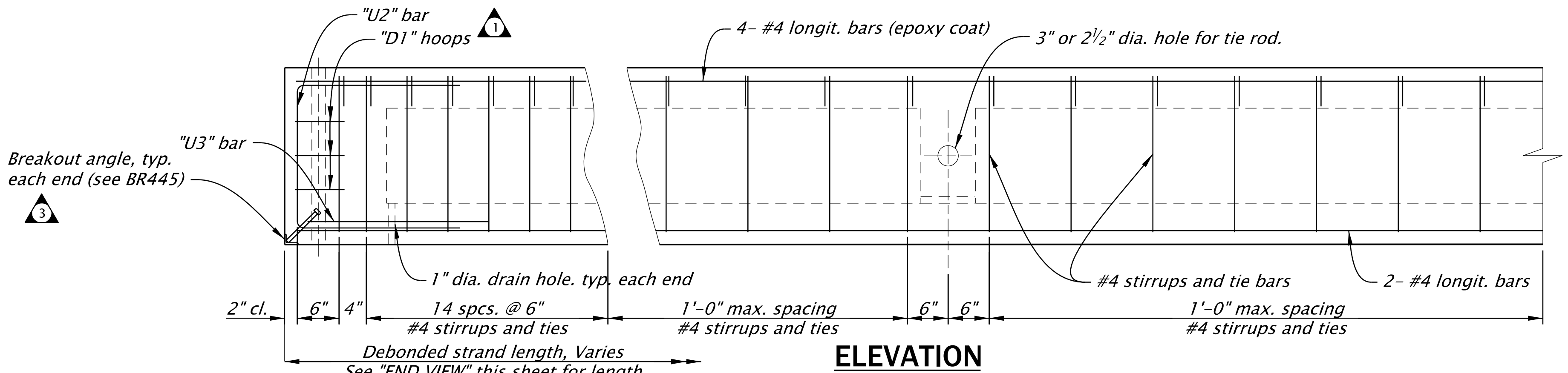
JUL-2022

BR420.dgn



PLAN

Scale: 1/4" = 1'-0"



ELEVATION

Scale: 1/2" = 1'-0"

NOTE "A" (Slab End Bars)
2- #4 x 10'-0", slabs 6 and 7
4- #4 x 12'-6", slabs 8, 9,10, 11, 12 and 13
Place bars each end of each slab (Epoxy coat).

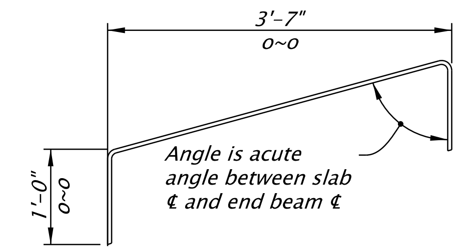
NOTE:
Grout keyway as specified in General Notes.
Omit keyway on exterior side of exterior slabs.
Keyway is continuous.

SECTION PROPERTIES

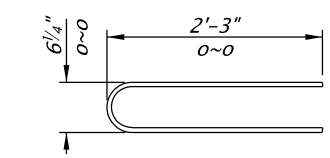
Area =	851in²
c.g. =	12.92 in
I =	63,596 in⁴
St =	4862in³
Sb =	4923in³
Weight =	916 lbs/ft
J =	128,800 in⁴
K =	0.77
V/S =	5.75
Form wt = (tubes)	20 lbs/ft
Total wt w/forms=	936 lbs/ft

Diaphragm Weight

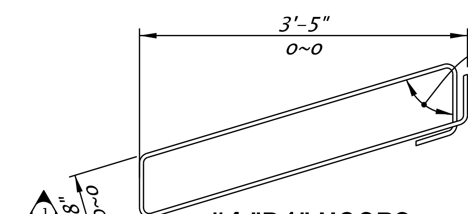
No Skew	430 lb
15° Skew	710 lb
30° Skew	1230 lb
45° Skew	1940 lb



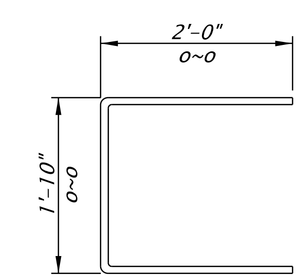
#4 "U1" BAR
(4- required per slab)



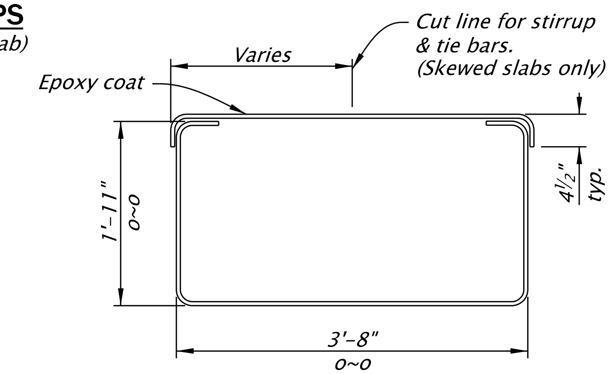
#6 "U3" BAR (A706)
(2- required per slab)



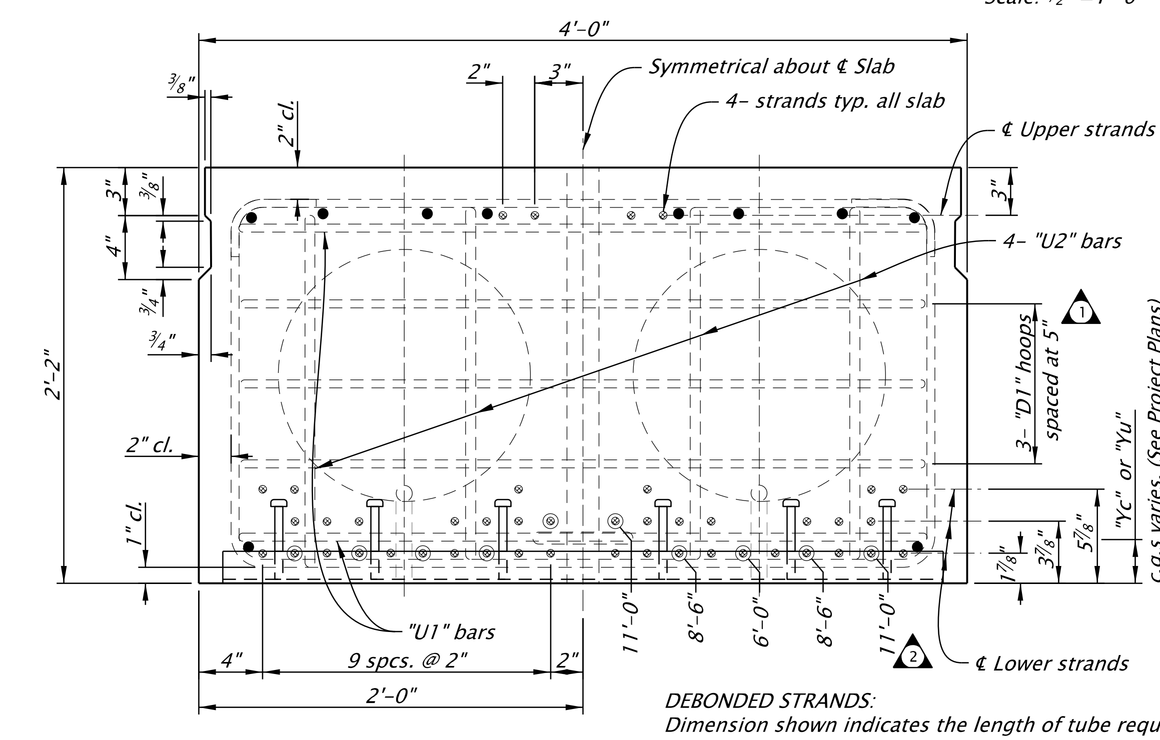
#4 "D1" HOOPS
(6- required per slab)



#5 "U2" BAR
(8- required per slab)



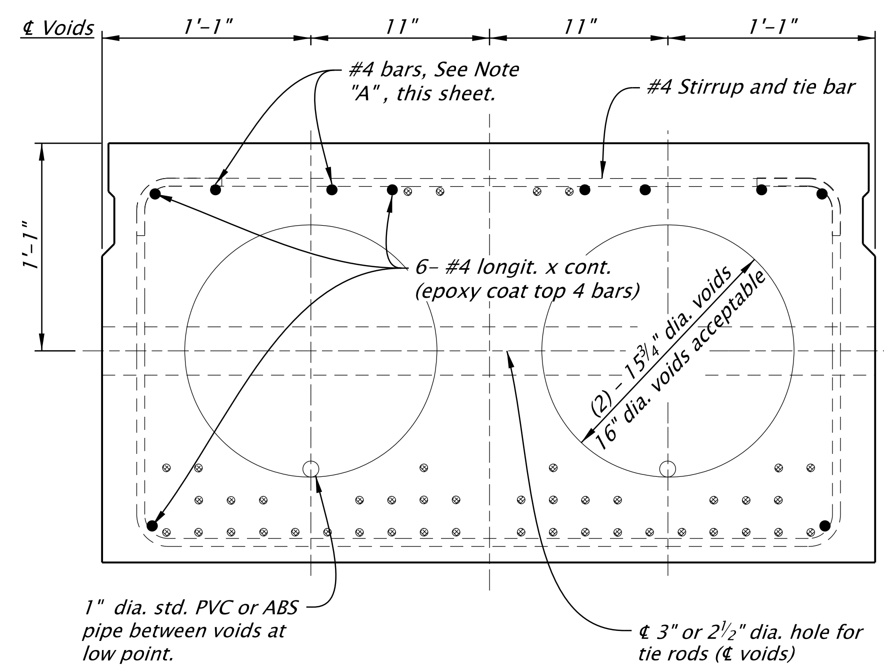
#4 STIRRUP & TIE



END VIEW

Scale: 1" = 1'-0"

DEBONDED STRANDS:
Dimension shown indicates the length of tube required, at both ends, to debond the indicated strand. See the "PRECAST PRESTRESSED SLAB SCHEDULE" in the project plans for the strand pattern detail.
Detension strands in order of increasing tube length shown.



MIDSPAN VIEW

Scale: 1" = 1'-0"

Accompanied by dwg. BR445

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All materials shall be in accordance with the current Oregon Standard Specifications.

OREGON STANDARD DRAWINGS

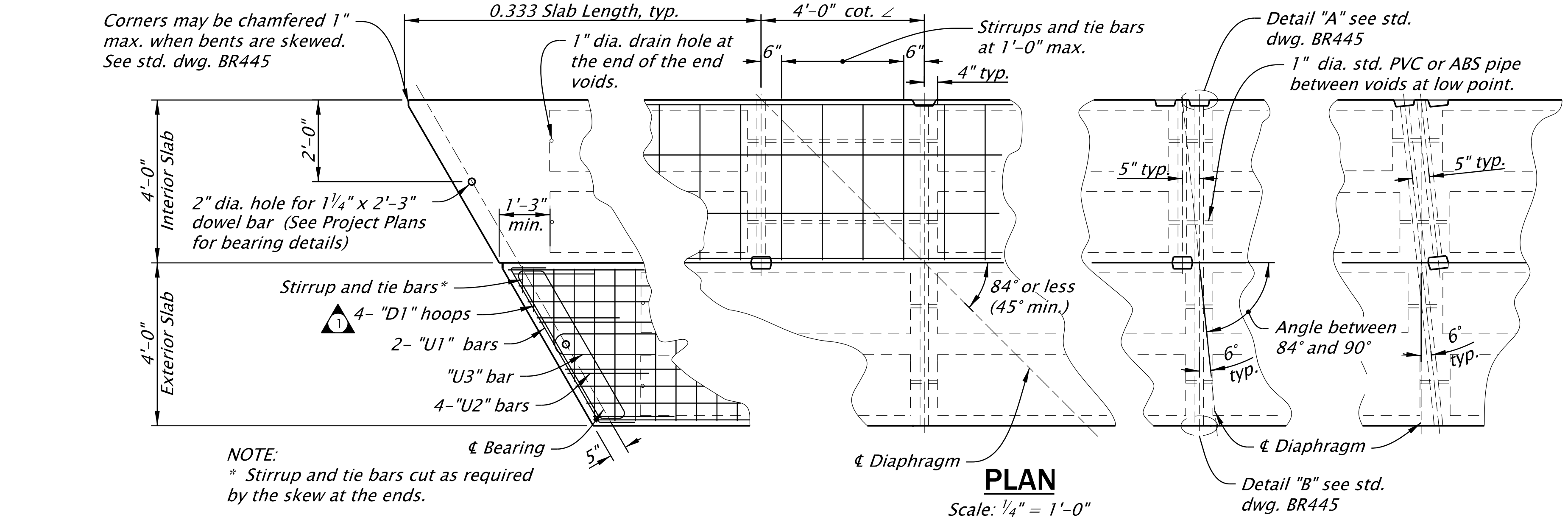
26" PRECAST PRESTRESSED SLAB

2024

	DATE	REVISION DESCRIPTION
1	07-2020	Added end zone reinf.
2	07-2020	Revised debonded length.
	07-2020	Updated drawing to current stds.
3	07-2022	Added breakout angle.

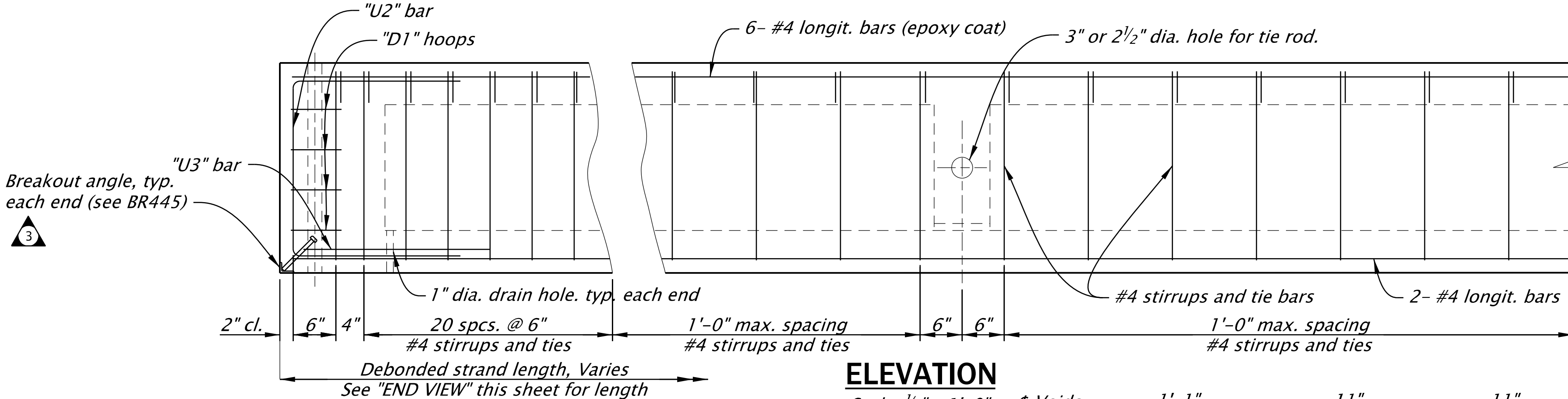
CALC. BOOK NO. N/A | SDR DATE 08-JUL-2022 | BR420

Effective Date: December 1, 2023 - May 31, 2024



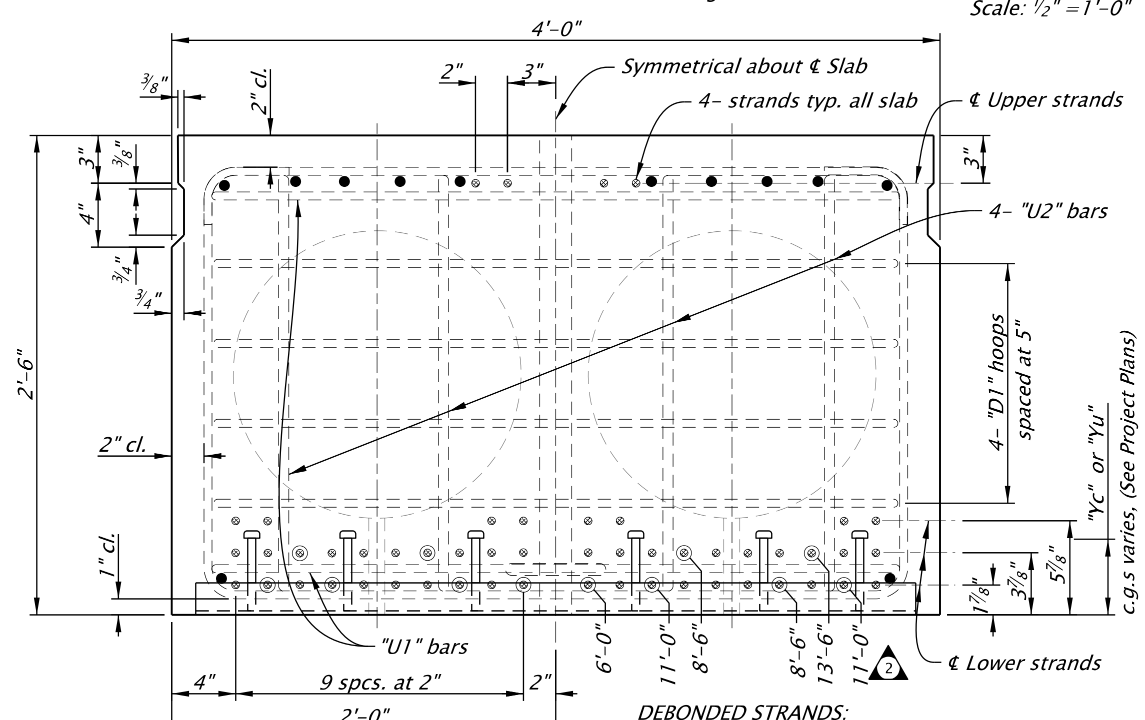
PLAN

Scale: ¼" = 1'-0"



ELEVATION

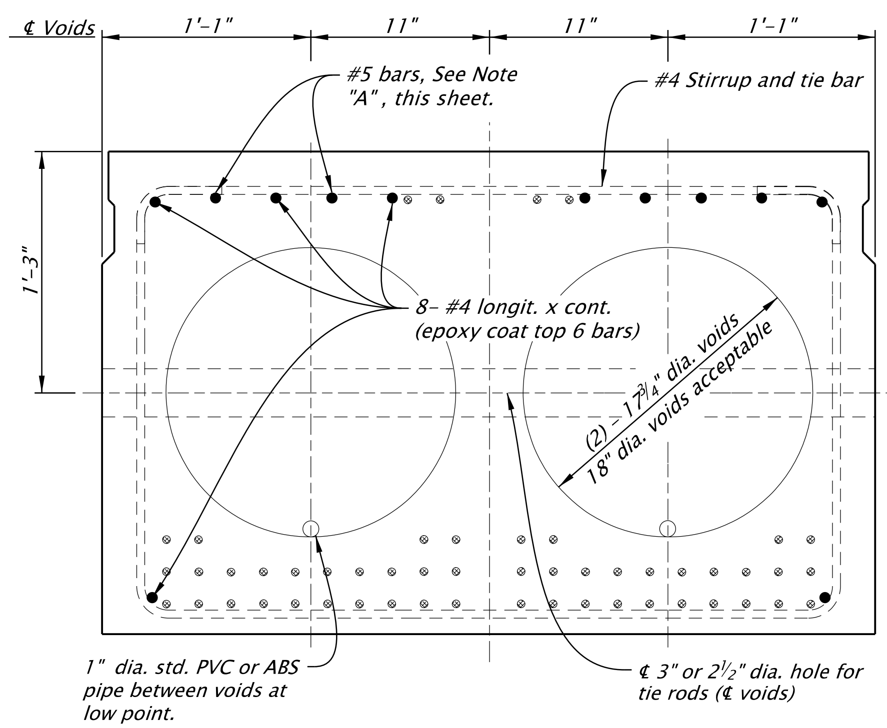
Scale: ½" = 1'-0"



END VIEW

Scale: 1" = 1'-0"

DEBONDED STRANDS:
Dimension shown indicates the length of tube required, at both ends, to debond the indicated strand. See the "PRECAST PRESTRESSED SLAB SCHEDULE" in the project plans for the strand pattern detail.
Detension strands in order of increasing tube length shown.



MIDSPAN VIEW

Scale: 1" = 1'-0"

NOTE "A" (Slab End Bars)
2- #5 x 10'-0", slabs 2 and 3
4- #5 x 10'-0", slabs 4, 5 and 6
4- #5 x 12'-6", slabs 7, 8 and 9
4- #5 x 15'-0", slabs 10, 11, 12 and 13
Place bars each end of each slab (Epoxy coat).

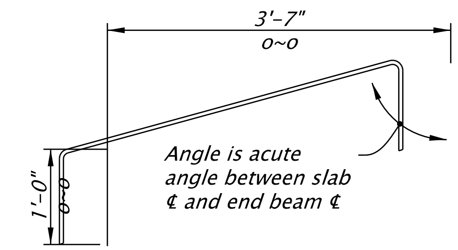
NOTE:
Grout keyway as specified in General Notes.
Omit keyway on exterior side of exterior slabs.
Keyway is continuous.

SECTION PROPERTIES

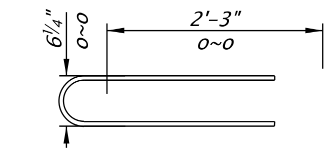
Area =	938 in^2
c.g. =	14.91 in
I =	97,278 in^4
St =	6447 in^3
Sb =	6524 in^3
Weight =	1009 lbs/ft
J =	180,800 in^4
K =	0.80
V/S =	6.01
Form wt = (tubes)	27 lbs/ft
Total wt w/forms=	1036 lbs/ft

Diaphragm Weight

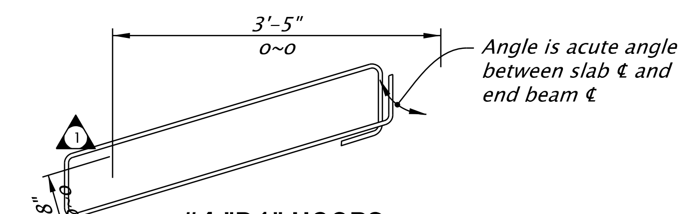
No Skew	550 lb
15° Skew	900 lb
30° Skew	1560 lb
45° Skew	2380 lb



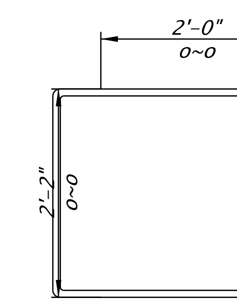
#4 "U1" BAR
(4- required per slab)



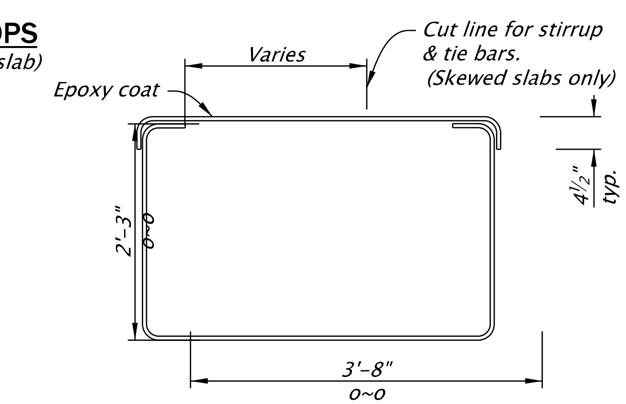
#6 "U3" BAR (A706)
(2- required per slab)



#4 "D1" HOOPS
(8- required per slab)



#5 "U2" BAR
(8- required per slab)



#4 STIRRUP & TIE

Accompanied by dwg. BR445

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All materials shall be in accordance with the current Oregon Standard Specifications.

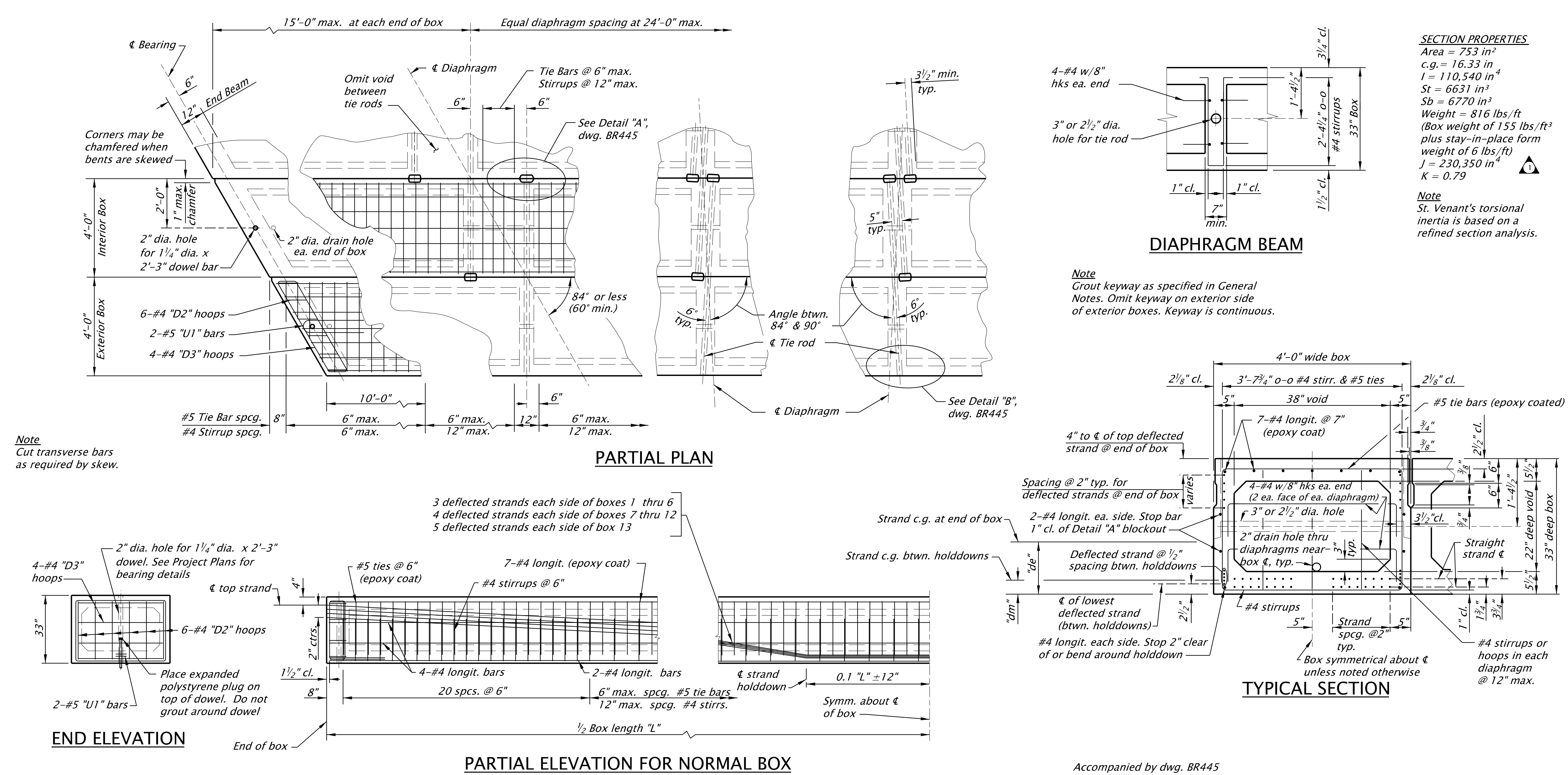
OREGON STANDARD DRAWINGS

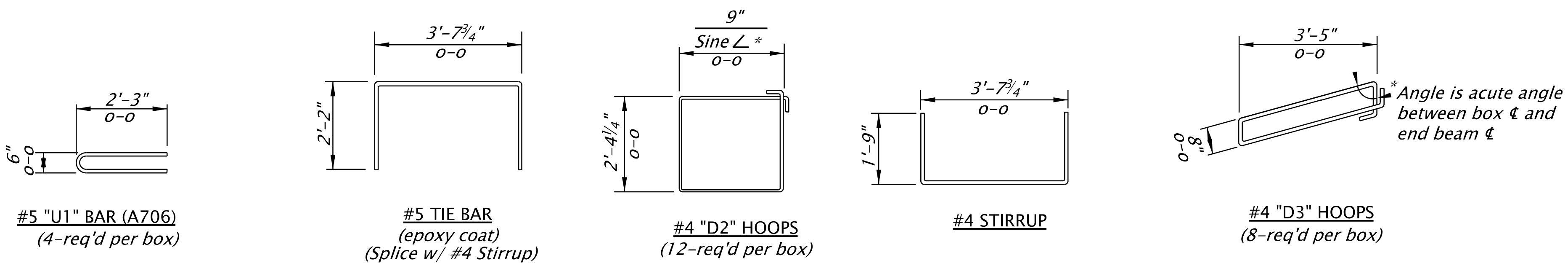
30" PRECAST PRESTRESSED SLAB

2024

	DATE	REVISION DESCRIPTION
2	07-2020	Added end zone reinf.
3	07-2020	Revised debonded lengths and locations
	07-2020	Updated drawing to current stds.
3	07-2022	Added breakout angle.

CALC. BOOK NO. N/A | SDR DATE 08-JUL-2022 | **BR422**





Accompanied by dwg. BR445

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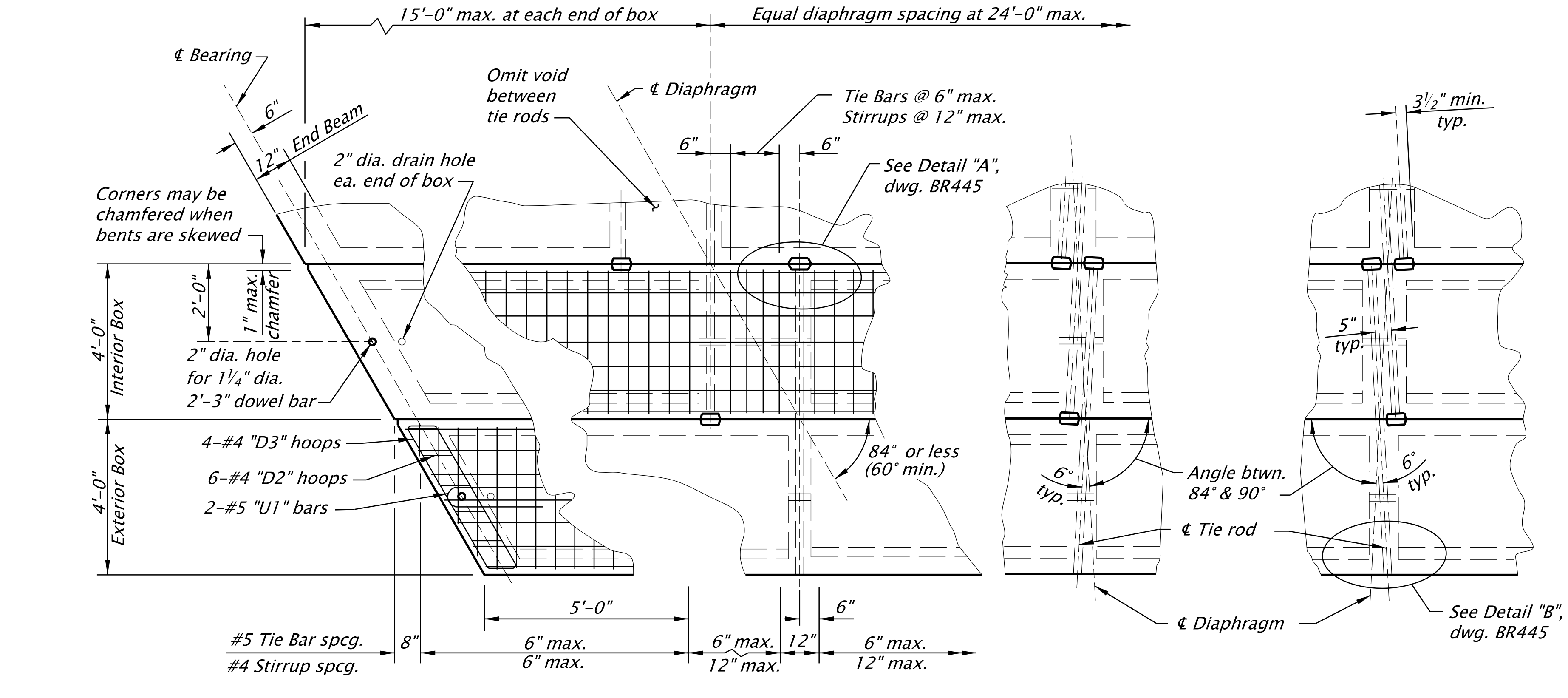
OREGON STANDARD DRAWINGS

33" PRECAST PRESTRESSED BOX

2024

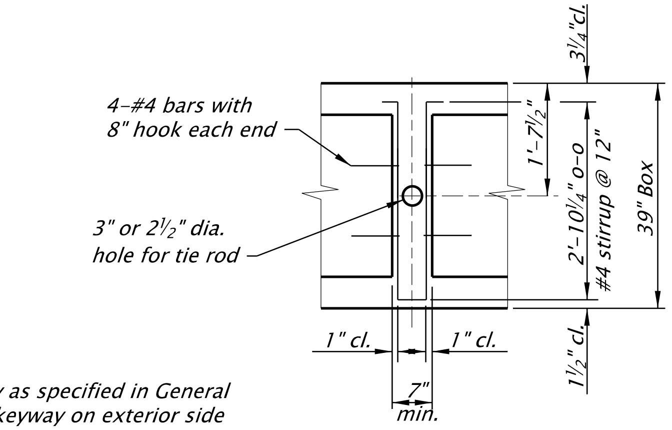
DATE	REVISION DESCRIPTION
-	-

CALC. BOOK NO. N/A | SDR DATE 20-APR-2018 | **BR425**



Note
Cut transverse bars as required by skew.

PARTIAL PLAN

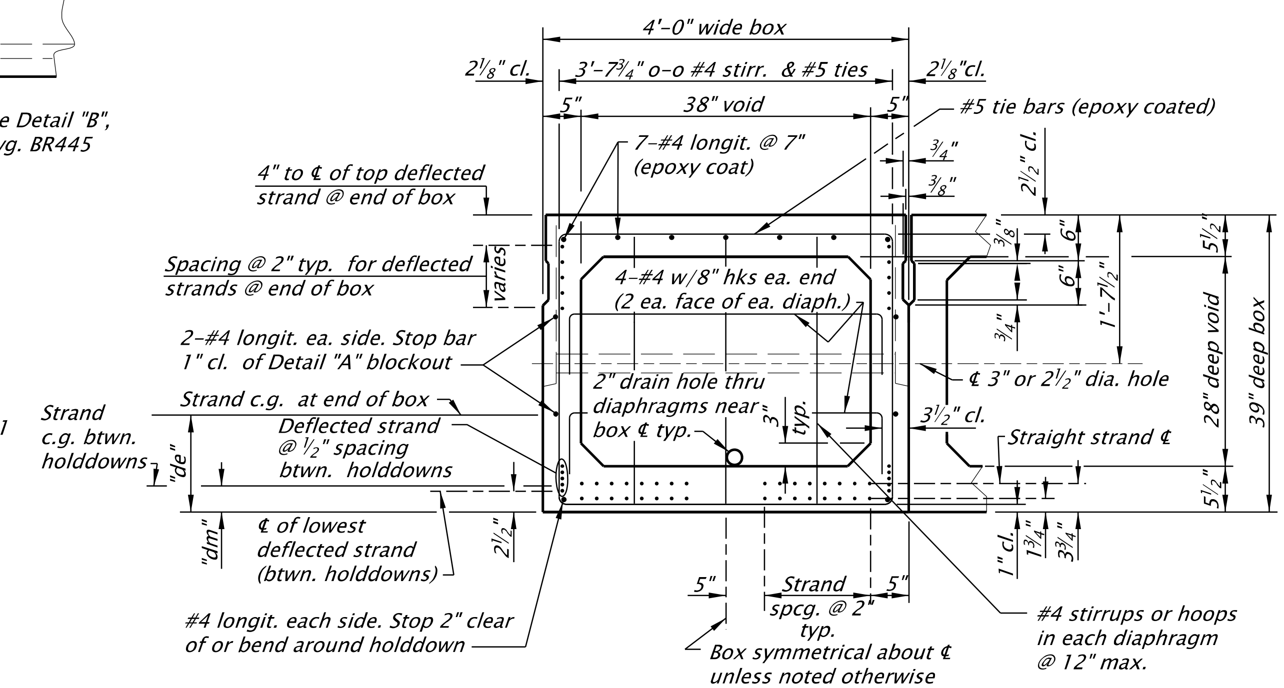


Note
Grout keyway as specified in General Notes. Omit keyway on exterior side of exterior boxes. Keyway is continuous.

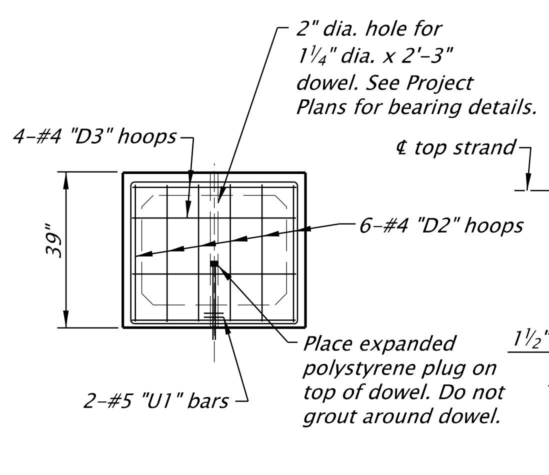
DIAPHRAGM BEAM

SECTION PROPERTIES
Area = 813 in²
c.g. = 19.29 in
I = 168,600 in⁴
St = 8555 in³
Sb = 8739 in³
Weight = 882lbs/ft
(Box weight of 155 lbs/ft³ plus stay-in-place form weight of 7 lbs/ft)
J = 311,500 in⁴
K = 0.84

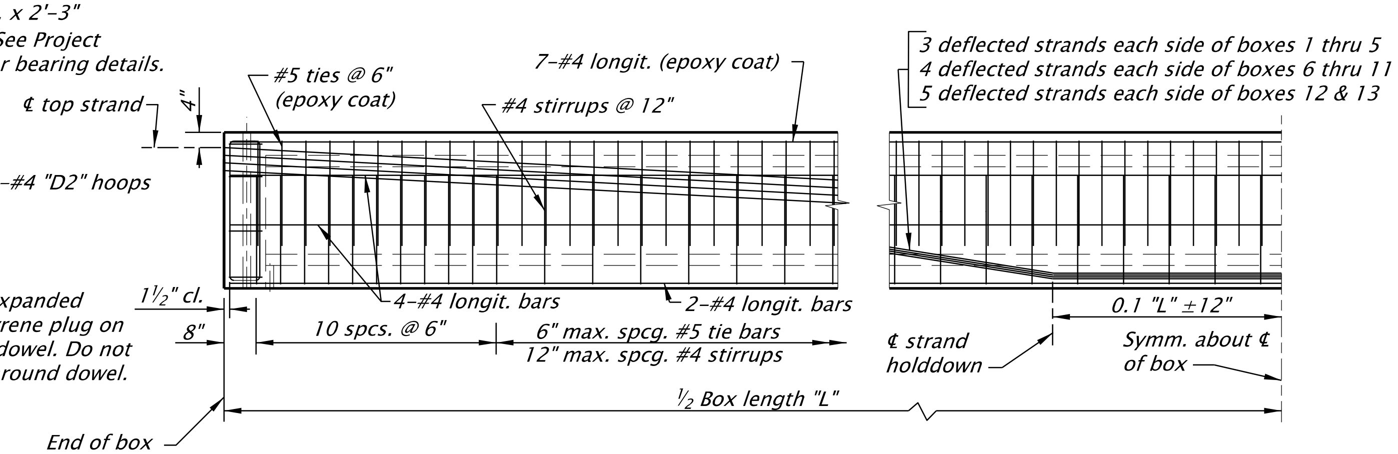
Note
St. Venant's torsional inertia is based on a refined section analysis.



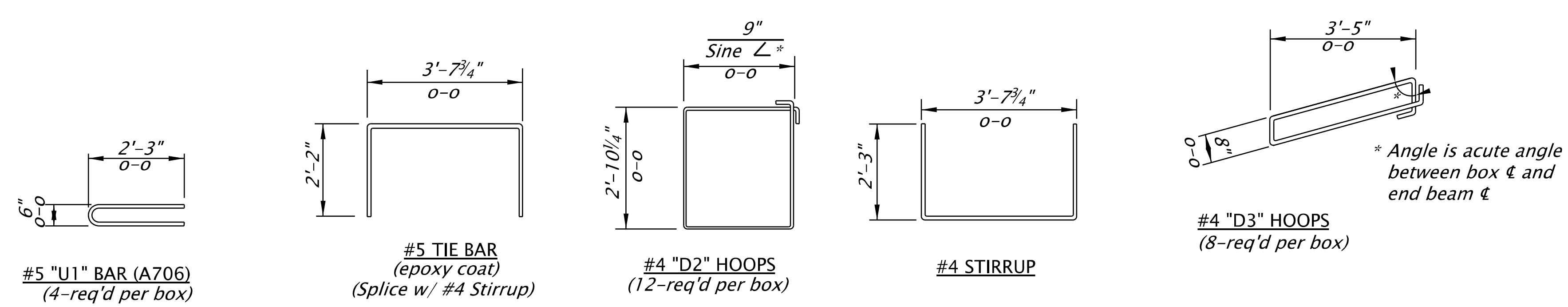
TYPICAL SECTION



END ELEVATION



PARTIAL ELEVATION FOR NORMAL BOX



Accompanied by dwg. BR445

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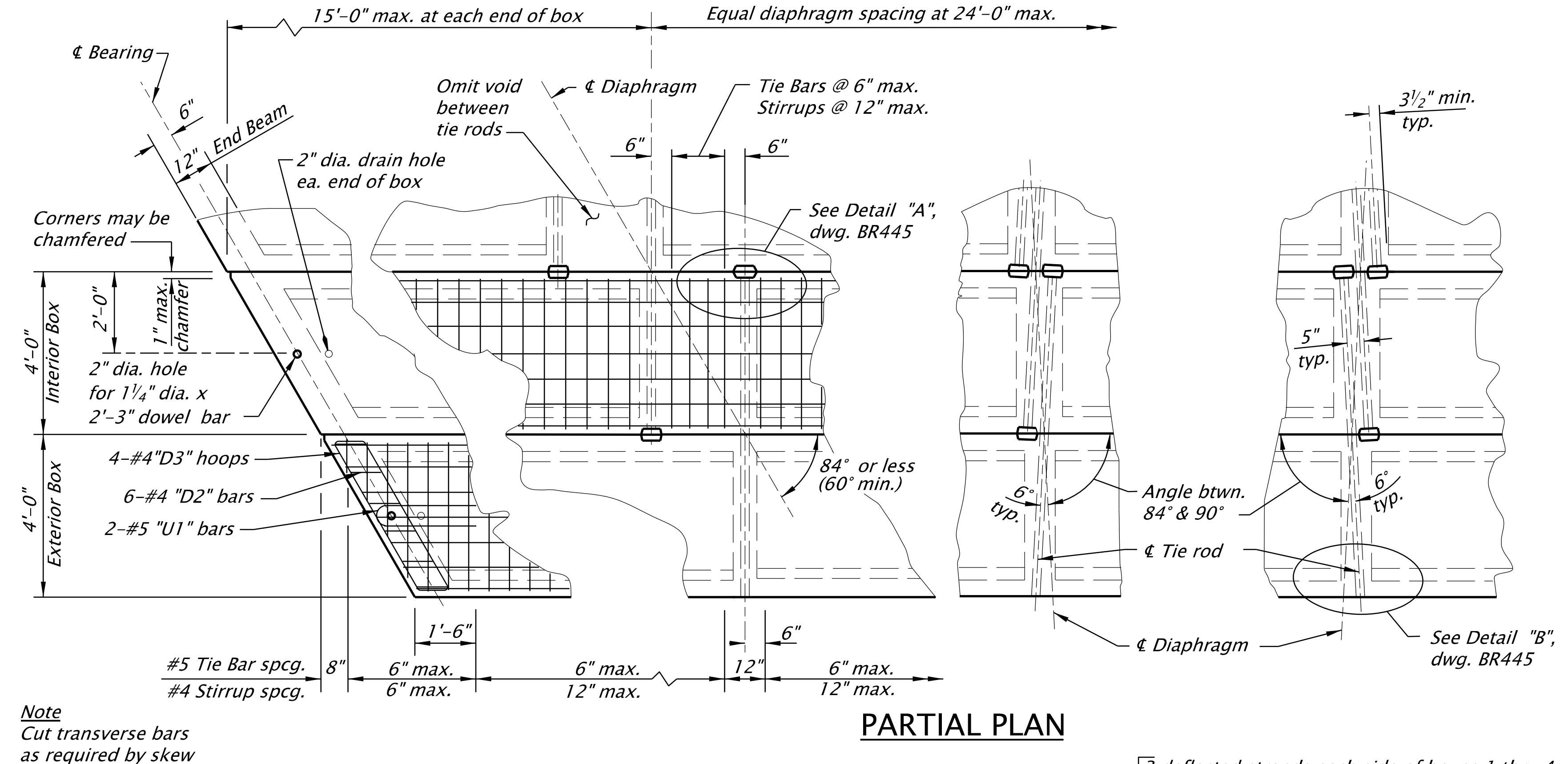
OREGON STANDARD DRAWINGS

39" PRECAST PRESTRESSED BOX

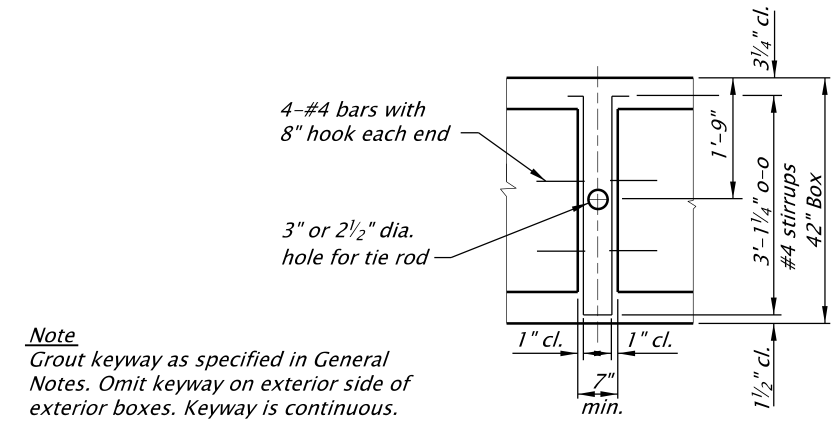
2024

DATE	REVISION DESCRIPTION
-	-

CALC. BOOK NO. N/A | SDR DATE 20-APR-2018 | BR430



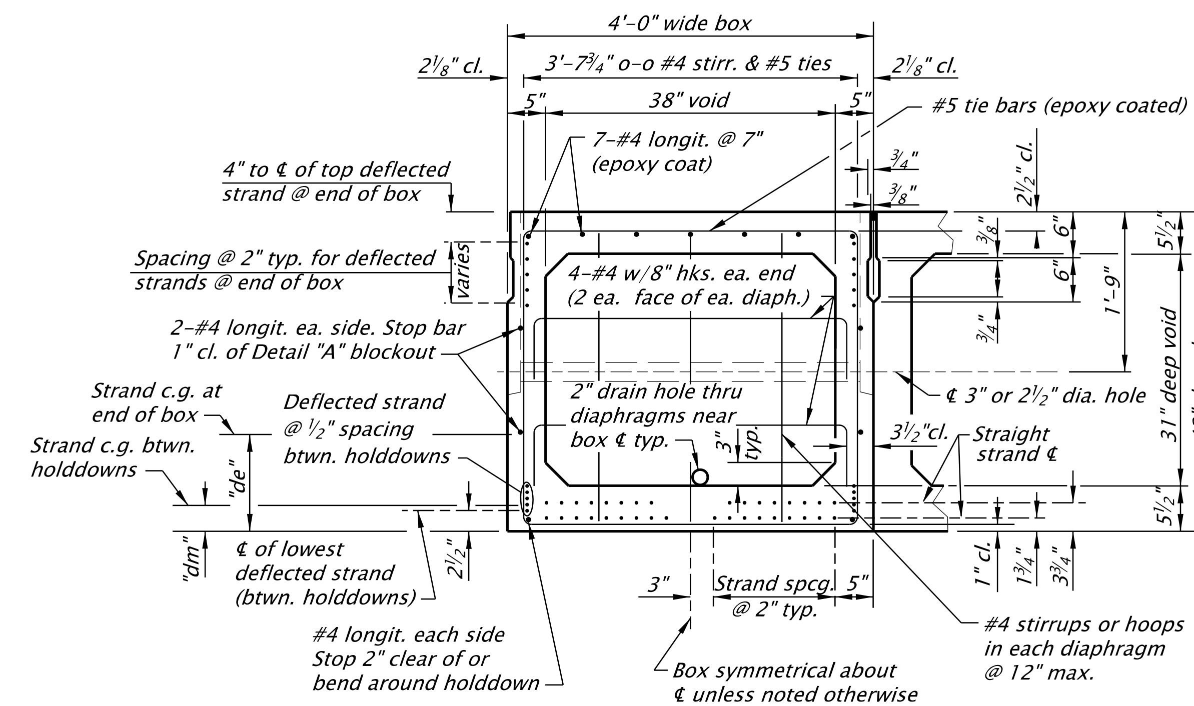
PARTIAL PLAN



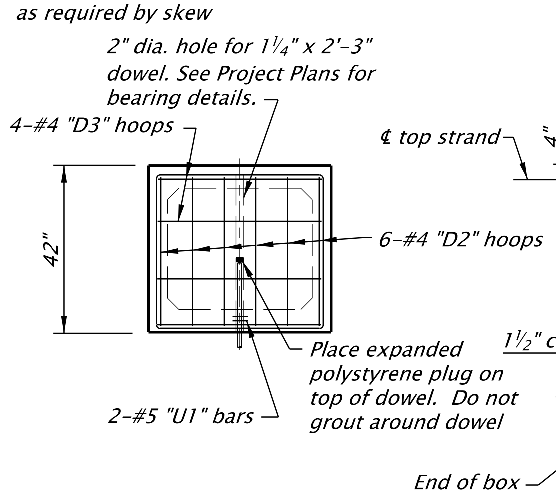
DIAPHRAGM BEAM

SECTION PROPERTIES
Area = 843 in^2
c.g. = 20.78 in
I = 203,050 in^4
St = 9567 in^3
Sb = 9772 in^3
Weight = 915 lbs/ft
(Box weight of 155 lbs/ft^3 plus stay-in-place form weight of 8 lbs/ft)
J = 354,400 in^4
K = 0.86

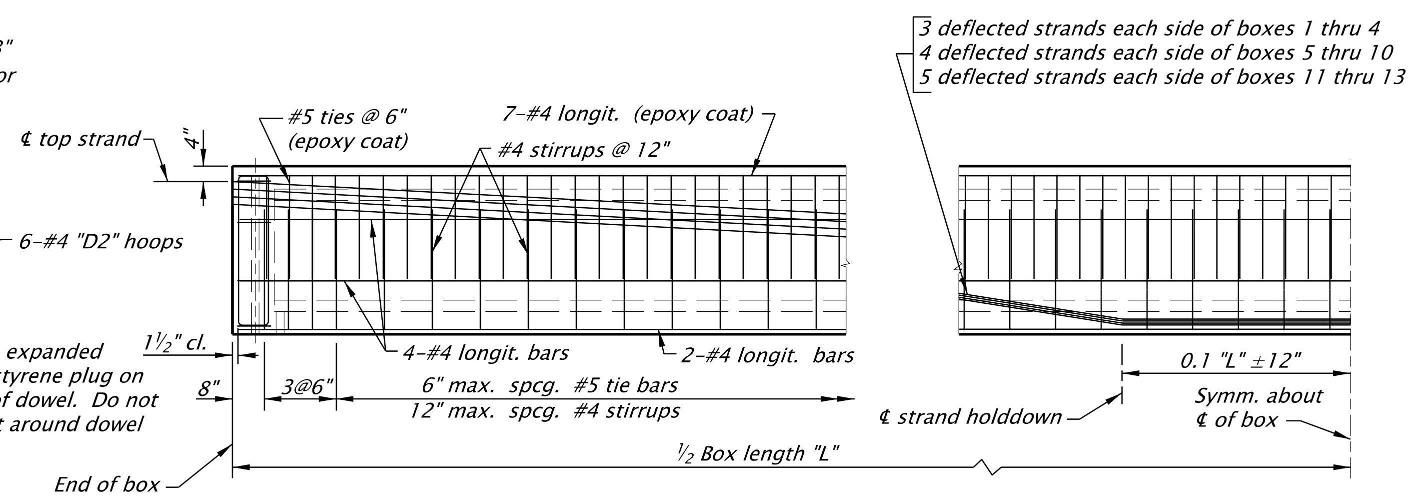
Note
St. Venant's torsional inertia is based on a refined section analysis.



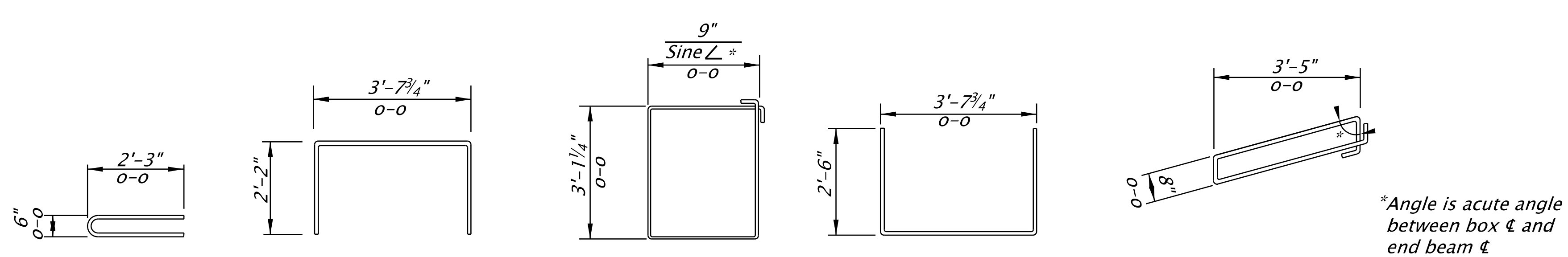
TYPICAL SECTION



END ELEVATION



PARTIAL ELEVATION FOR NORMAL BOX



#5 "U1" BAR (A706)
(4-req'd per box)

#5 TIE BAR
(epoxy coat)
(Splice w/ #4 Stirrup)

#4 "D2" HOOPS
(12-req'd per box)

#4 STIRRUP

#4 "D3" HOOPS
(8-req'd per box)

Accompanied by dwg. BR445

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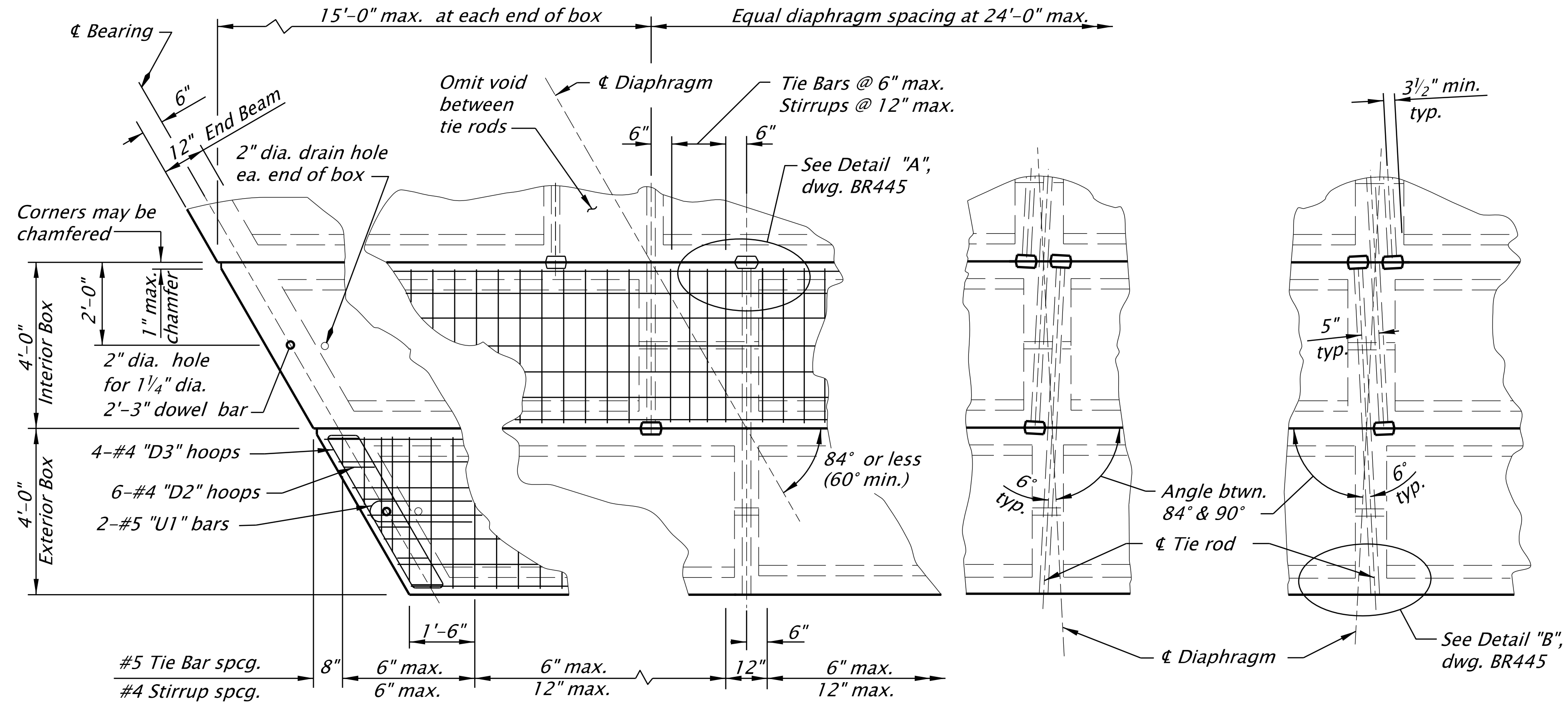
OREGON STANDARD DRAWINGS

42" PRECAST PRESTRESSED BOX

2024

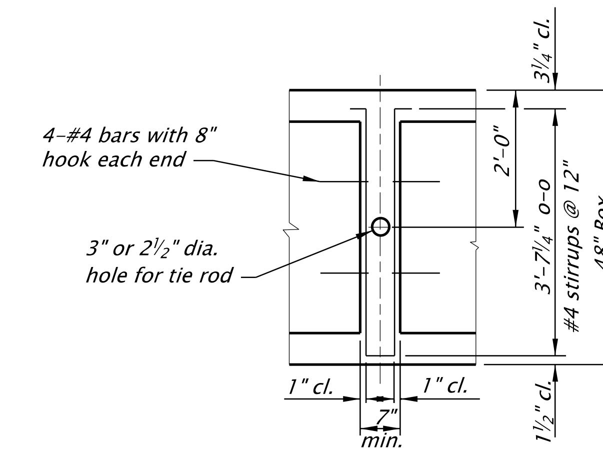
DATE	REVISION DESCRIPTION
-	-

CALC. BOOK NO. N/A | SDR DATE 20-APR-2018 | **BR435**



PARTIAL PLAN

Note
Cut transverse bars as required by skew.

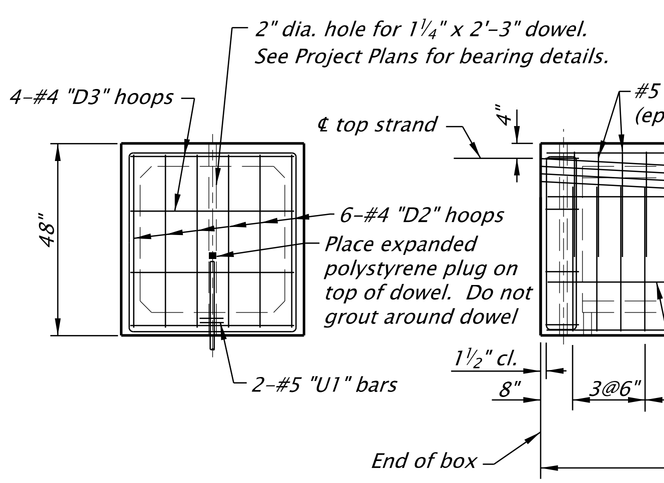


DIAPHRAGM BEAM

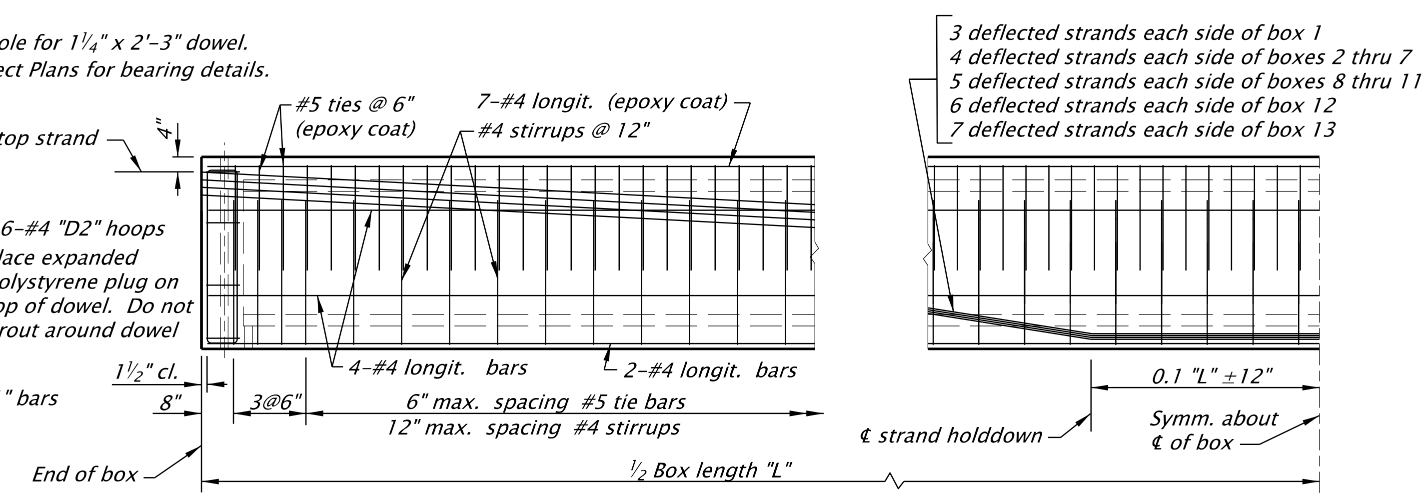
SECTION PROPERTIES
Area = 903 in²
c.g. = 23.75 in
I = 283,450 in⁴
St = 11,688 in³
Sb = 11,936 in³
Weight = 982 lbs/ft
(Box weight of 155 lbs/ft plus stay-in-place form weight of 8 lbs/ft)
J = 442,850 in⁴
K = 0.91

Note
St. Venant's torsional inertia is based on a refined section analysis.

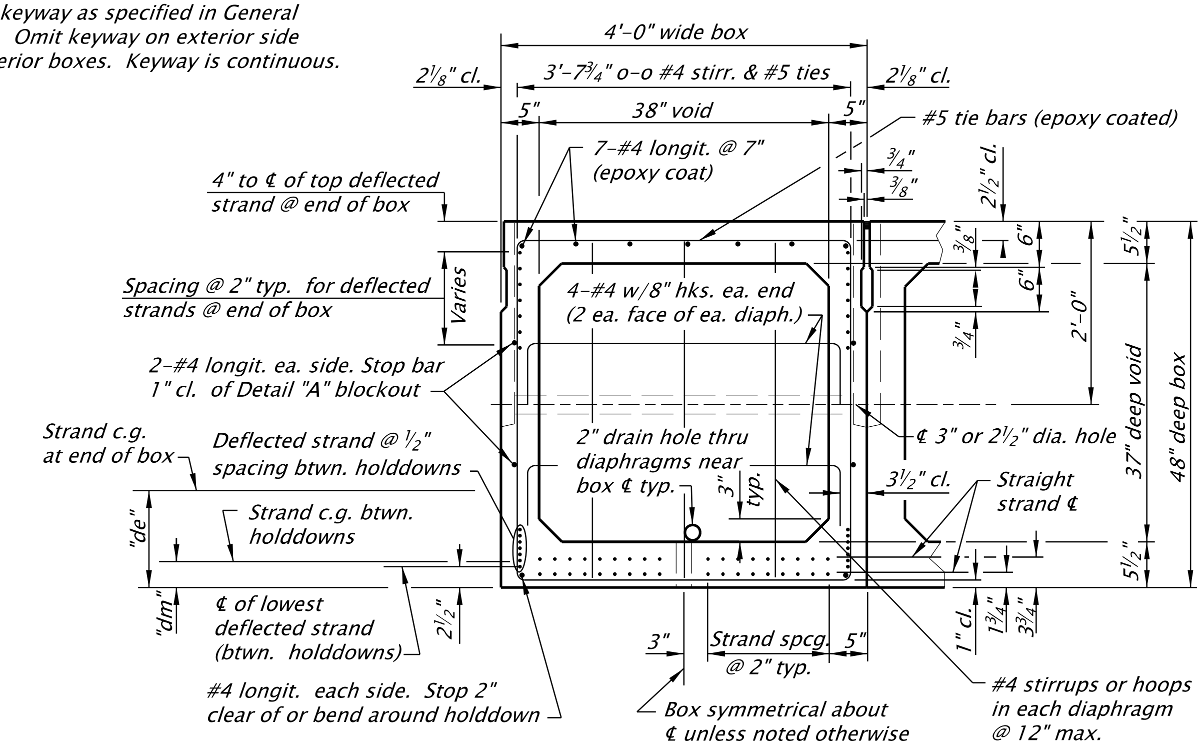
Note
Grout keyway as specified in General Notes. Omit keyway on exterior side of exterior boxes. Keyway is continuous.



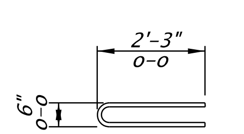
END ELEVATION



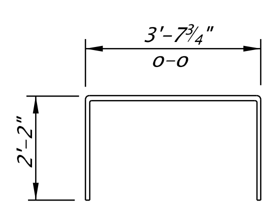
PARTIAL ELEVATION FOR NORMAL BOX



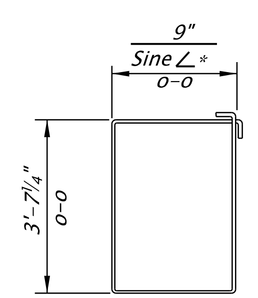
TYPICAL SECTION



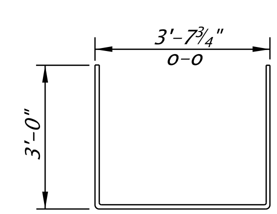
#5 "U1" BAR (A706)
(4-req'd per box)



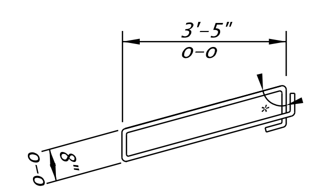
#5 TIE BAR
(epoxy coat)
(Splice w/ #4 Stirrup)



#4 "D2" HOOPS
(12-req'd per box)



#4 STIRRUP



* Angle is acute angle between box ¢ and end beam ¢

#4 "D3" HOOPS
(8-req'd per box)

Accompanied by dwg. BR445

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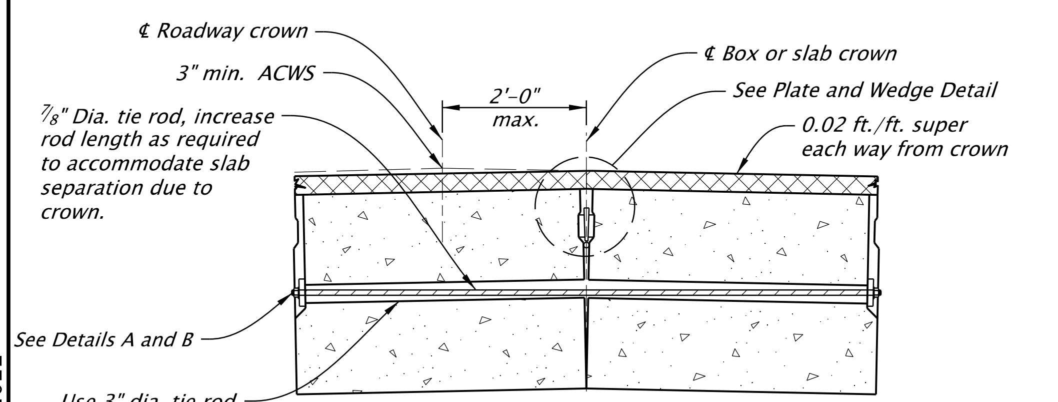
OREGON STANDARD DRAWINGS

48" PRECAST PRESTRESSED BOX

2024

DATE	REVISION DESCRIPTION
-	-

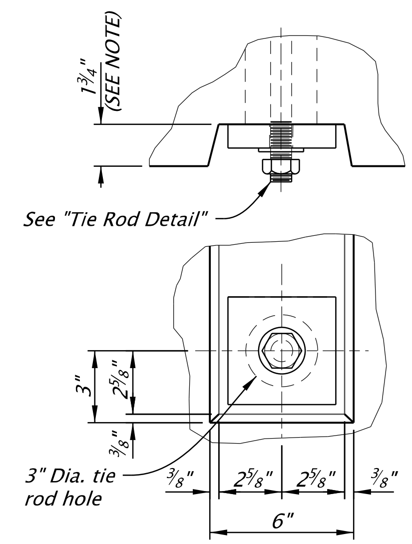
CALC. BOOK NO. N/A — SDR DATE 20-APR-2018 — BR440



TYPICAL DETAIL FOR INSTALLING BOXES OR SLABS ON CROWN

NOTE:
Tighten tie rods until the bottom corners of the boxes or slabs are in contact. Loosen the tie rod and install the plates and wedges per detail. Shift wedge location as required to avoid conflict with the tie rod.

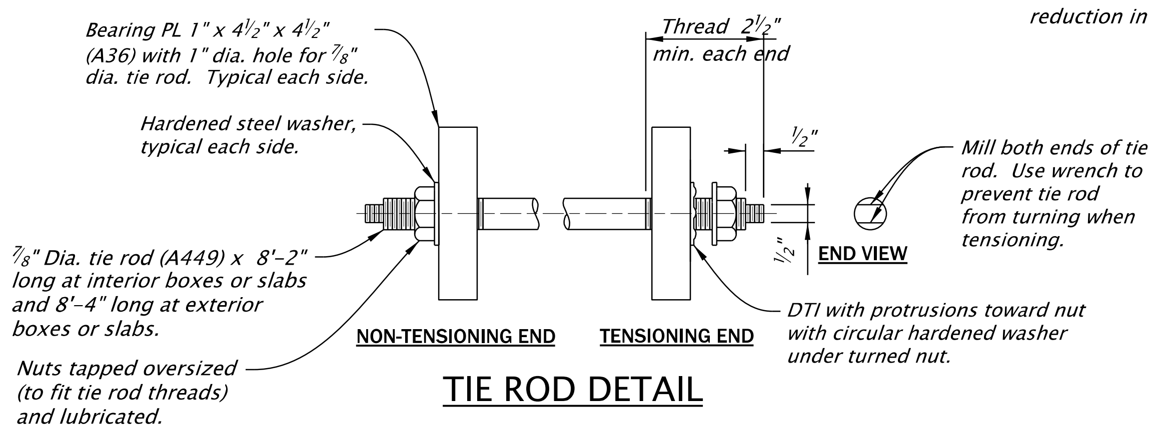
Tension the tie rods. Install boxes or slabs level and build up roadway crown with AC wearing surface when roadway width is 28' or less (for bridges with ACWS).



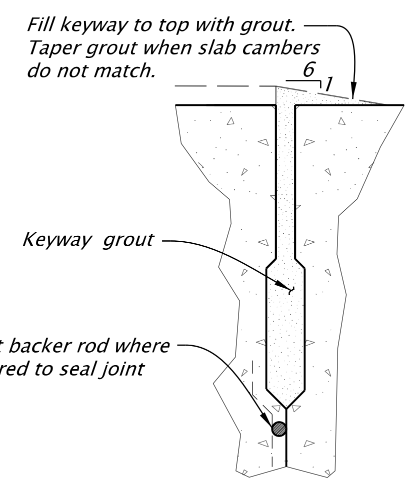
DETAIL "A"

Non-Tensioning End

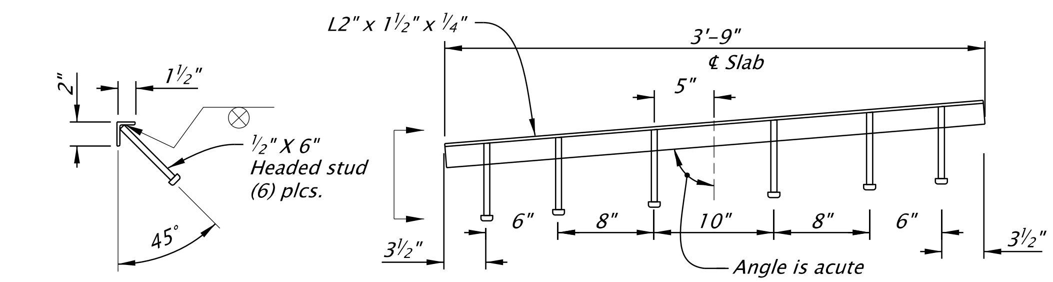
NOTE:
1 3/4" @ tie rod ¢ (2 1/2" depth may be used for slabs with an appropriate reduction in tie rod length)



TIE ROD DETAIL

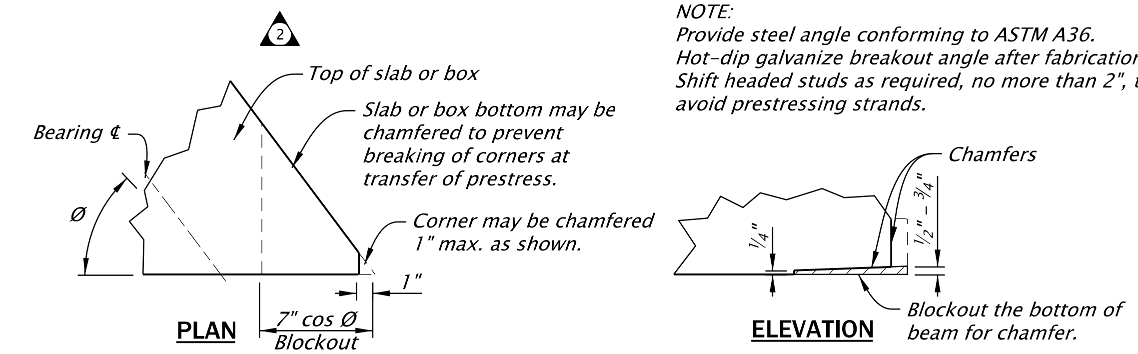


KEYWAY GROUT DETAIL



BREAKOUT ANGLE DETAIL

NOTE:
Provide steel angle conforming to ASTM A36.
Hot-dip galvanize breakout angle after fabrication.
Shift headed studs as required, no more than 2", to avoid prestressing strands.



PARTIAL ELEVATION CHAMFER DETAIL

GENERAL NOTES FOR PRESTRESSED BOXES AND SLABS

Boxes and slabs are designed for live and superimposed dead loading as shown in the General Notes for the Project.

Provide the class of concrete shown in the Slab or Box Schedule with nominal maximum size aggregate of 1 or 3/4. Transfer prestress after the concrete reaches the minimum concrete strength at transfer shown in the Slab or Box Schedule.

Select a keyway grout from the QPL for filling keyways, lifting blockouts and tie rod blockouts.

Allow traffic on the bridge only after keyway grout has reached design strength.

Provide reinforcing steel as specified in the General Notes for the Project.

Provide smooth dowels conforming to AASHTO M31, Grade 60 (ASTM A615, Grade 60), ASTM F1554, Grade 55 or ASTM A529, Grade 55. (2) (1)

Provide 1/2" diameter 7 wire low relaxation prestressing steel strand conforming to AASHTO Specification M203 (ASTM A416), Grade 270 Supplement 1.

Tension strand initially to 31.0 kips per strand (after harping deflected strand). Debond strands where specified using either split or solid plastic sheathing with a minimum wall thickness of 0.025".

Provide high strength tie rods conforming to ASTM A449. Provide heavy hexagon nuts conforming to ASTM A563. Provide hardened steel washers conforming to ASTM F436. Hot-dip galvanize tie rods, plates, nuts and washers (except DTIs) after fabrication.

(1) Tighten tie rods to 39 kips (minimum) using mechanically galvanized direct tension indicators (DTIs) conforming to ASTM F959 and ASTM F3125, Grade A325. (1) Tighten all tie rods (per box or slab) to about one half the specified tension before proceeding with final tensioning.

Keep boxes and slabs upright at all times. Support them within 2'-0" of the ends during storage (to prevent excessive camber, overstress or failure). Locate transport supports and lifting devices within 2'-0" of the ends of boxes and slabs. Transportboxes and slabs after the concrete has reached the 28 day design strength and a minimum of 7 days after casting.

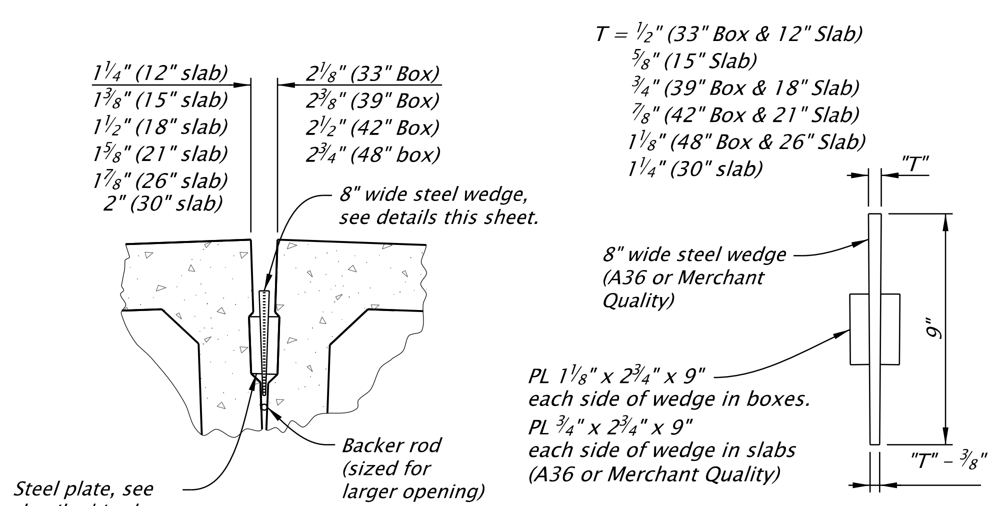
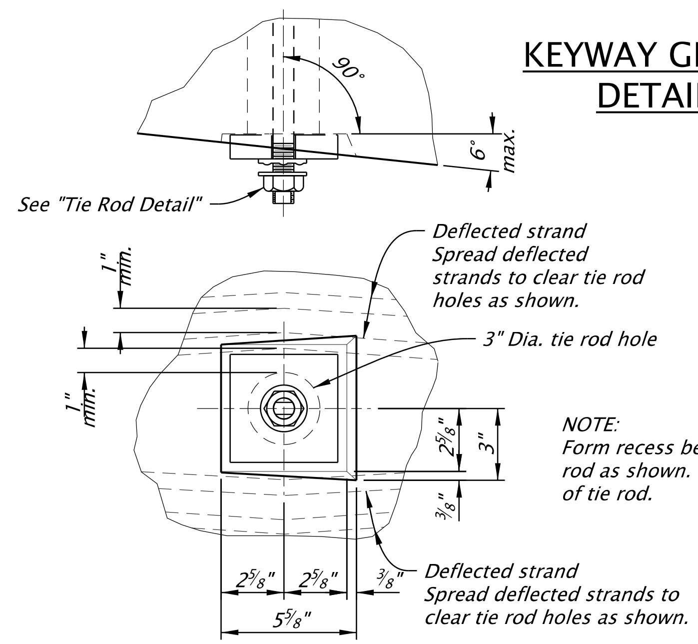


PLATE AND WEDGE DETAIL

NOTE:
Add steel plates and wedge at each tie rod crossing before tensioning tie rods.
Hot-dip galvanize wedges and plates.



DETAIL "B"

Tensioning End

NOTE:
Form recess bearing area perpendicular to tie rod as shown. Use Detail "B" at tensioning end of tie rod.

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OREGON STANDARD DRAWINGS

PRECAST PRESTRESSED BOX AND SLAB DETAILS

2024

	DATE	REVISION DESCRIPTION
4	07-2020	Revised steel grade for dowels and tie rods.
3	07-2020	Updated drawing to current stds.
2	07-2020	Moved rail anchorage detail to det3465.
1	06-2022	Revised General Notes. Added breakout Angle Detail.

CALC. BOOK NO. N/A | SDR DATE 08-JUL-2022 | BR445

JUN-2022

BR445.dgn

Effective Date: December 1, 2023 - May 31, 2024